



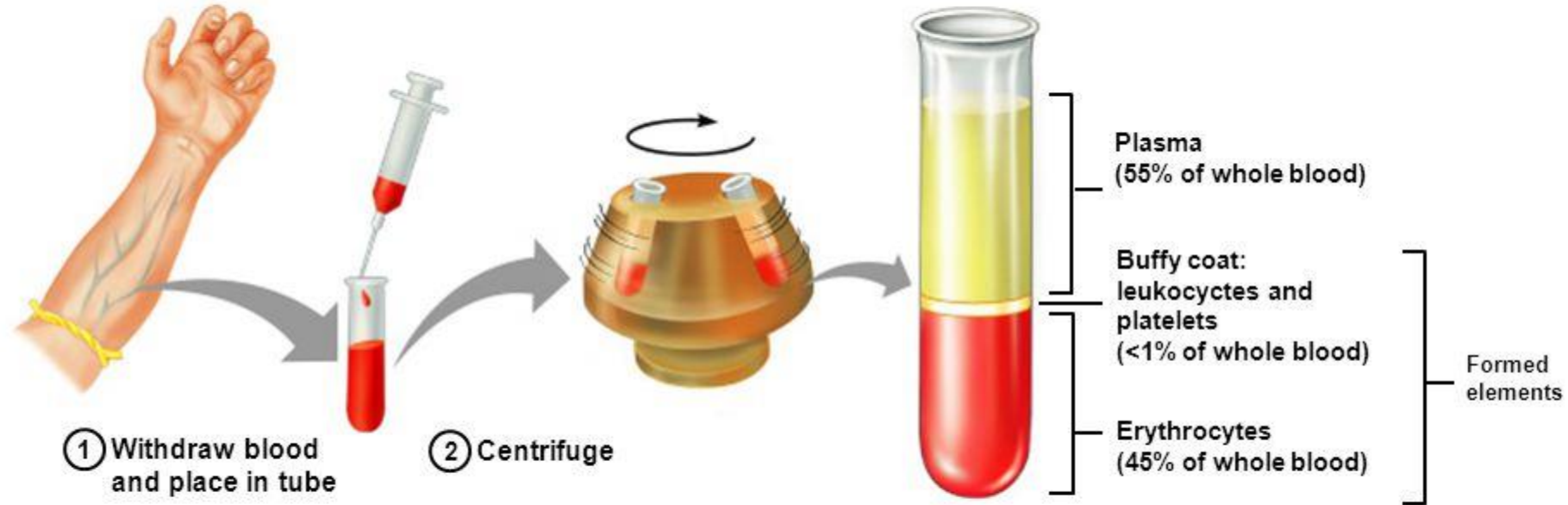
Erythrocytes

Structure, function, and metabolism

Dr. Jehad Alhmoud



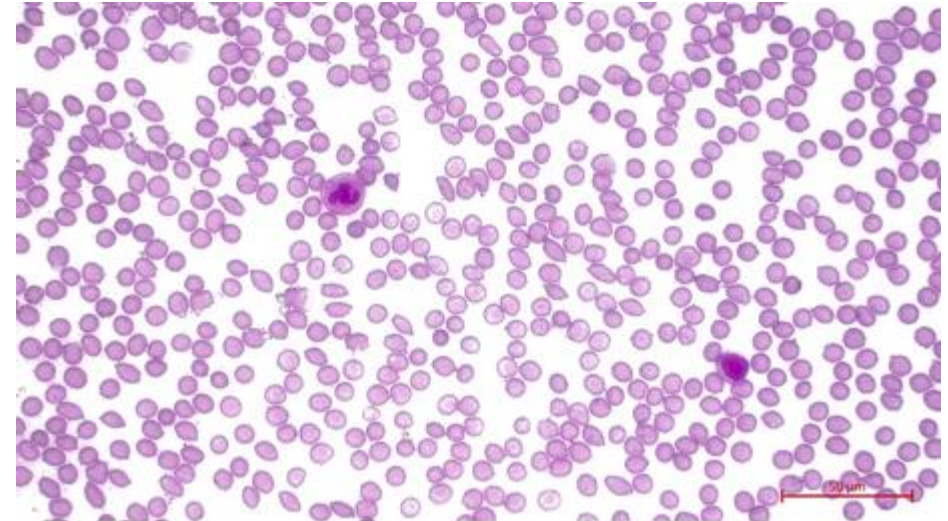
Components of Whole Blood



- Hematocrit
 - Males: $47\% \pm 5\%$
 - Females: $42\% \pm 5\%$

Erythrocytes

- 1) Biconcave disc.
- 2) They lack a cell nucleus (anucleate) and most organelles.
- 3) Have a diameter of 7-8 μm and a thickness of 2-2.5 μm at the periphery and 0.8-1.0 μm at the center.
- 4) Biconcave flexible shape to be able to enter 3.5 μm in diameter micro-vessels.
- 5) Have a life span of 120 days.
- 6) Function in O_2 transport from the lungs to the cells and CO_2 from the cells to the lungs to be exhaled.

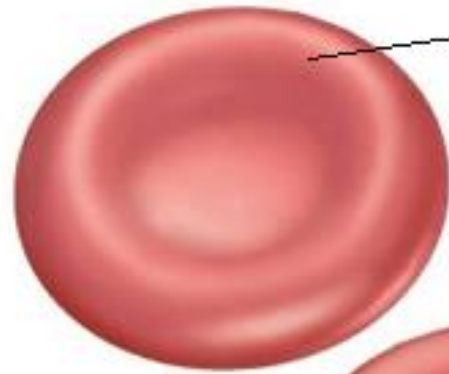


- 7) Contain iron-containing protein called hemoglobin.
- 8) Produced through a process called erythropoiesis.
- 9) The most abundant or numerous formed elements.
- 10) Responsible for the red color of the blood.
- 11) Highly capable of significant deformability.
- 12) The average volume (MCV) is 90 (80-100) fL (femto liter = 10^{-15}).

13) The average surface area is $140 \mu\text{m}^2$, a 40% excess of surface area compared with a 90-fL sphere.

14) This excess surface-to-volume ratio enables RBCs to stretch undamaged up to 2.5 times their resting diameter as they pass through narrow capillaries and splenic pores $2 \mu\text{m}$ in diameter; this property is called RBC deformability.

Red blood cells



cytoplasm
containing
haemoglobin



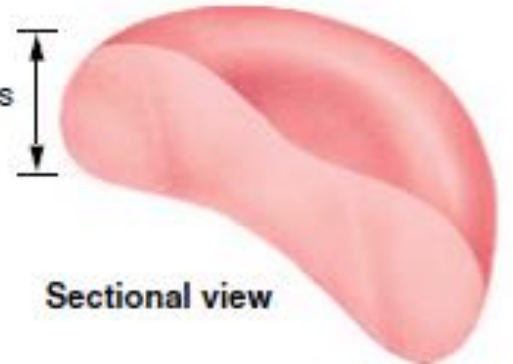
biconcave discs with no nucleus, carry oxygen

Top view



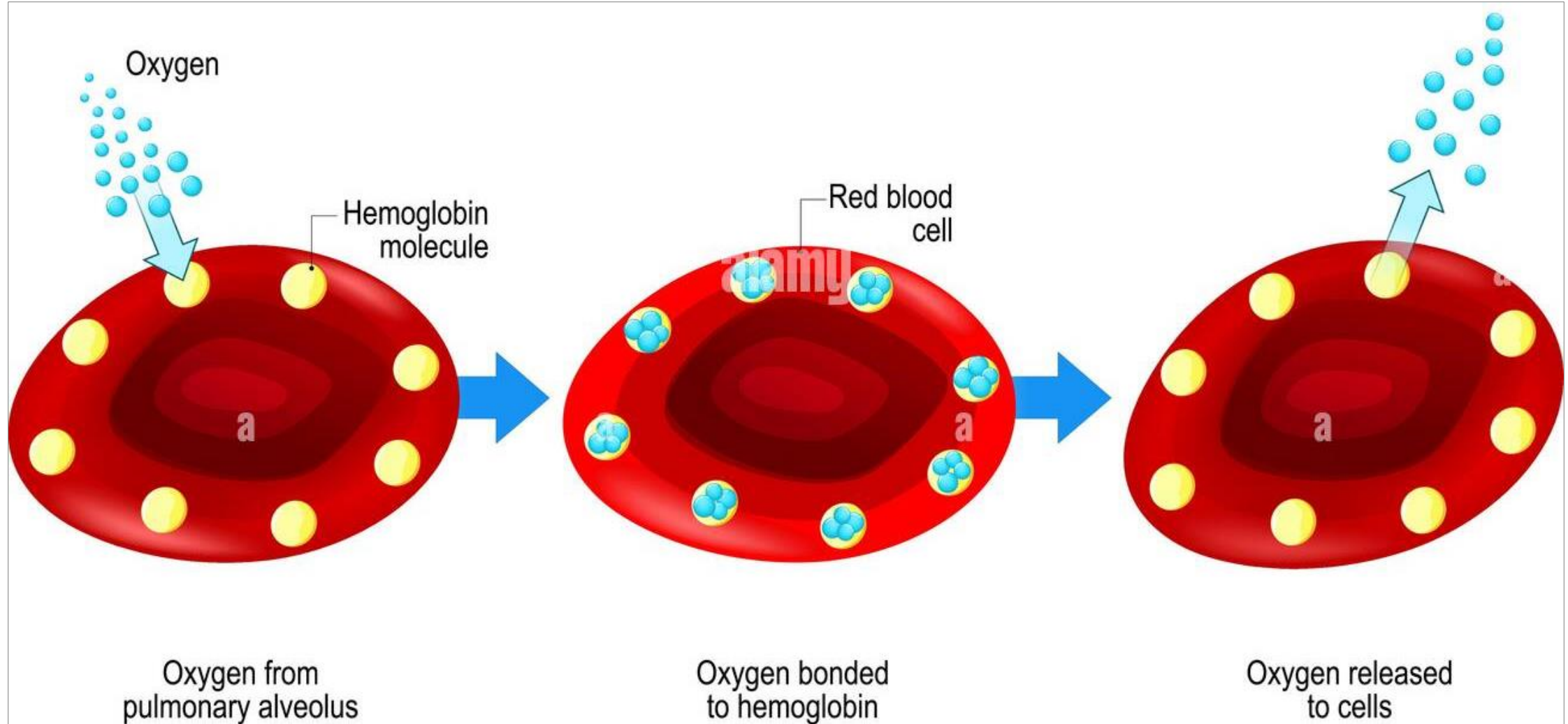
7.5 micrometers

2.0 micrometers



Sectional view

RBC transporting Oxygen Molecules Bonded with Hemoglobin



- Erythrocytes, or red blood cells, are simple cells that perform an essential function: the exchange of respiratory gasses.

- These cells are faced with several challenges.
 - 1) They lack nuclei, ribosomes, and other cellular organelles
 - 2) Have no protein synthesis or repair capability
 - 3) They carry oxygen (O_2), a toxic substance, but cannot use it in the process of energy generation like other cells of the body
 - 4) They must maintain a high internal concentration of potassium and a low internal concentration of sodium, against the concentration gradient for both ions, which requires energy.
 - 5) They must be flexible to squeeze through small capillaries and must also withstand the stresses of blood flow at high arterial pressures.

Assessment of erythropoiesis

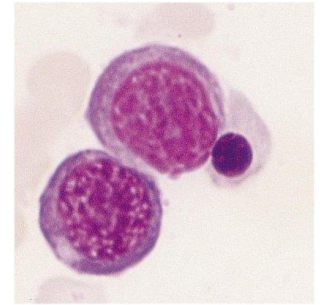
- Information needed
 1. bone marrow examination
 2. haemoglobin level
 3. reticulocyte count
- In bone marrow we look at myeloid: erythroid ratio ex.(granulocyte precursor: red cell precursor ratio) 2.5:1 to 12:1
- The ratio falls when erythropoiesis increases.
- Also, reticulocyte count increases in proportion to the degree of anemia.
- Ineffective erythropoiesis indicates for high ratio and low reticulocyte count.

RBC Membrane

- The RBC plasma membrane, which is 5 μm thick, is 100 times more elastic than a comparable latex membrane, yet it has tensile (lateral) strength greater than that of steel.
- The deformable RBC membrane provides the broad surface area and close tissue contact necessary to support the delivery of O_2 from lungs to body tissue and CO_2 from body tissue to lungs.

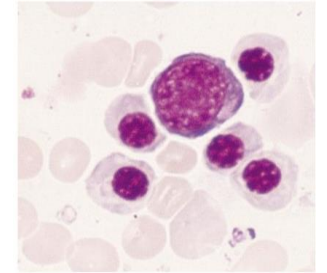
Erythropoiesis:

- A process of erythrocytes formation (RBCs).
- 10^{12} new RBCs are made every day in the body.
- Pass from Stem cell----GEMM-CFU----BFU-E----CFU-E----Pronormoblast----normoblast (early, intermediate and late)---reticulocytes---RBCs.



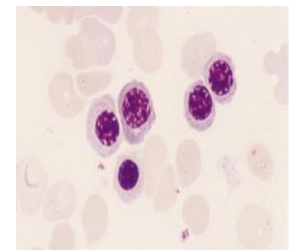
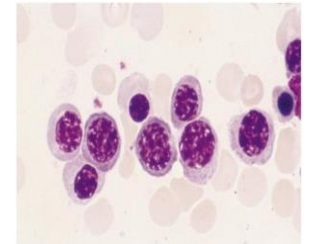
Pronormoblast:

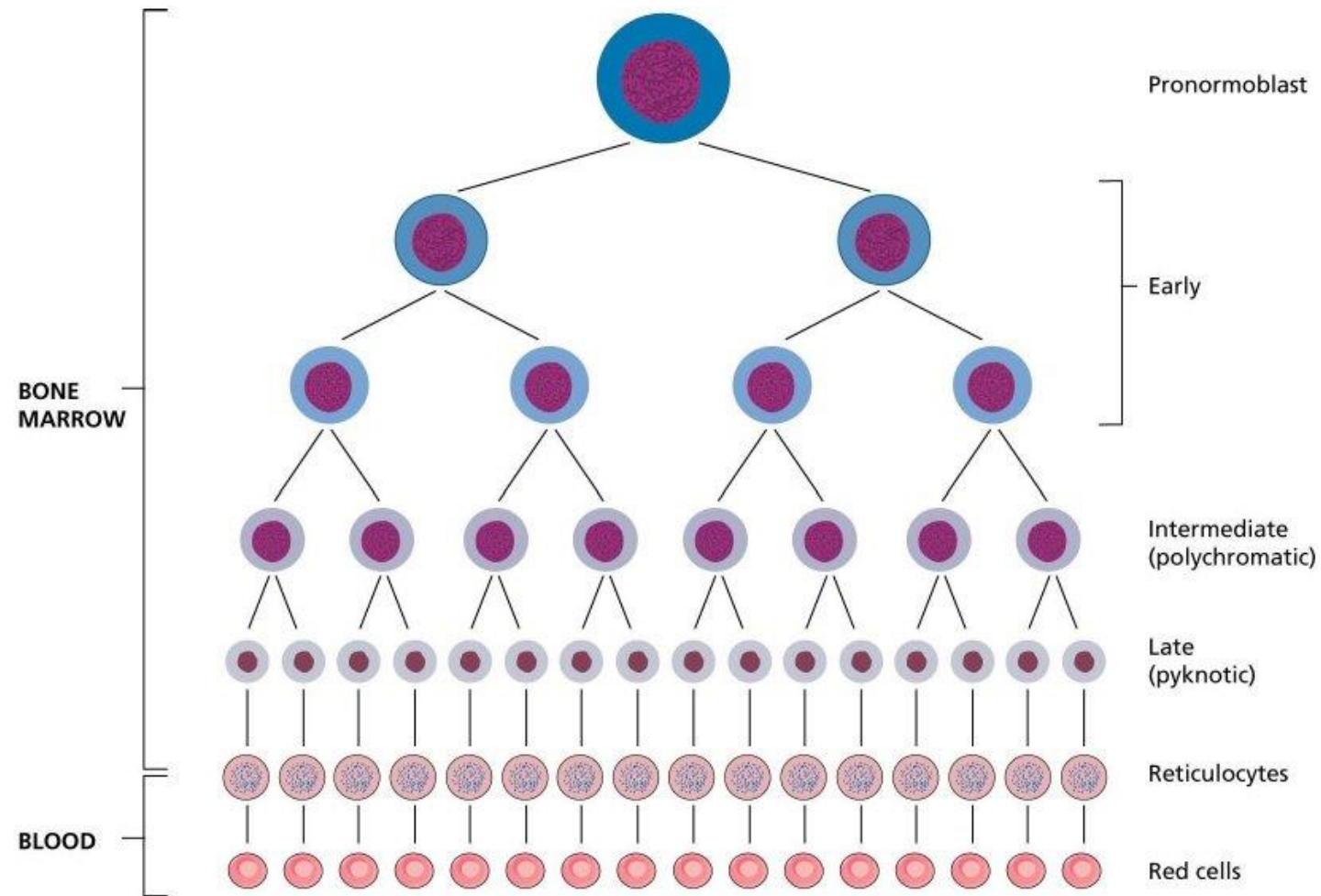
- 1st recognizable erythrocyte precursor
- Large cell with dark blue cytoplasm.
- Dark purple nucleolus with nuclei and slightly clumped chromatin
- Single Pronormoblast gives rise to 16 mature RBSs.



Normoblast:

- Raised by numbers of cell division.
- Get smaller down on the synthesis line.
- Stain pink or paler blue in the cytoplasm (progressive hemoglobin synthesis) and it loses RNA and protein synthesis apparatus.
- Nuclear chromatin became more dense.





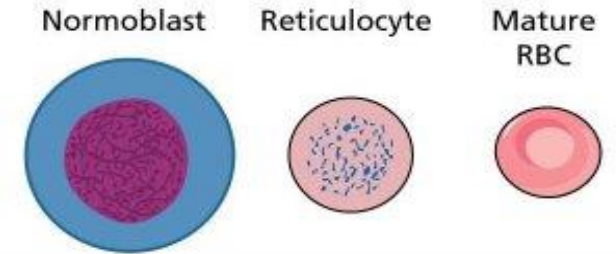
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Reticulocytes:

- Larger than mature RBC.
- Contains ribosomal RNA and is still able to synthesize hemoglobin.
- 1-2 days in the marrow and 1-2 days in the circulation.
- Convert to mature RBC when RNA is completely lost.

Mature RBCs:

- Completely pink staining.
- Non-nucleated, no RNA, and biconcave disk in shape.
- Function to carry O₂ to the tissue and return CO₂ to the lungs.

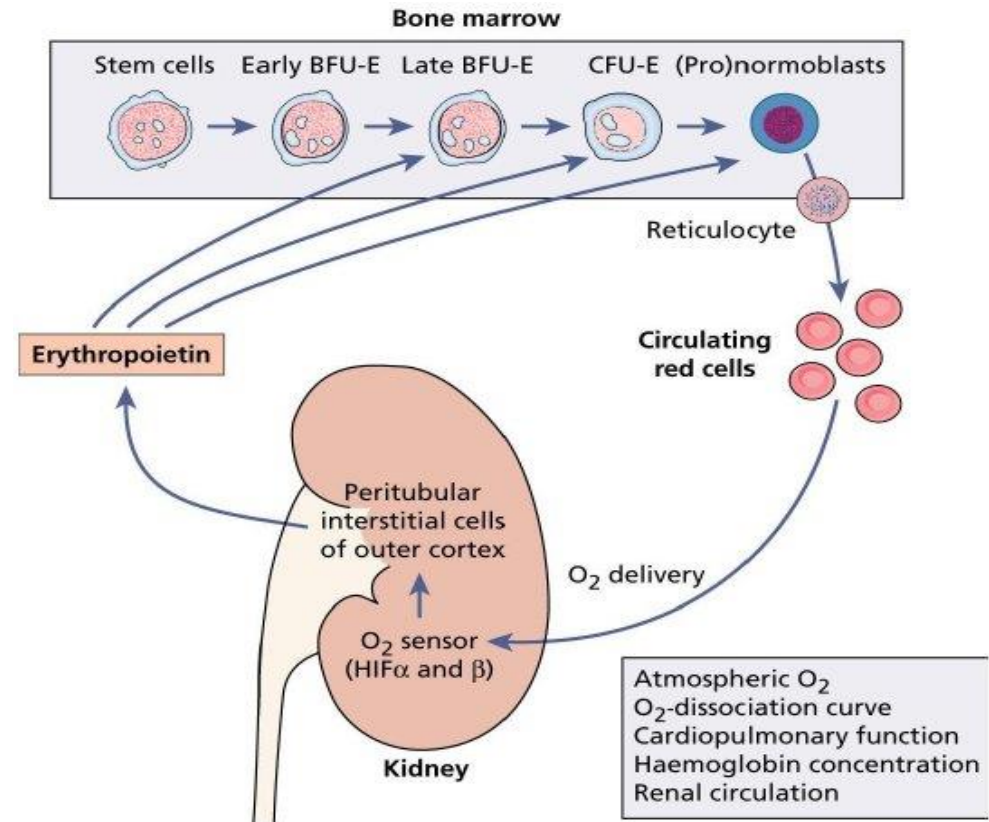


Nuclear DNA	Yes	No	No
RNA in cytoplasm	Yes	Yes	No
In marrow	Yes	Yes	Yes
In blood	No	Yes	Yes

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Erythropoietin

- A hormone that regulates RBC formation or erythropoiesis.
- Glycosylated polypeptide of 165 amino acids and 34Kd in MW.
- 90% is produced in the kidney and 10% in the liver.
- Stimulated by oxygen tension in the kidney.
- Hypoxia induces hypoxia-inducible factors (HIF-2 α and β)
- These factors act on the HIF response element on the erythropoietin gene and switch on its production.
- The hormone increases RBC synthesis by acting on its progenitor cells committed to erythrocyte formation.
- The hormone acts on GATA-1 and FOG-1 transcription factors of the progenitor cells to enhance the expression of erythroid-specific genes (haem and red cell membrane synthesis).
- Increase O₂ reduces erythropoietin synthesis.



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Erythropoietin clinical uses:

- Anemia of chronic renal diseases.
- Myelodysplastic syndrome.
- Anemia associated with malignancy and chemotherapy.
- Anemia of chronic disease e.g. Rheumatoid arthritis (RA).
- Anemia of prematurity.
- Perioperative uses.

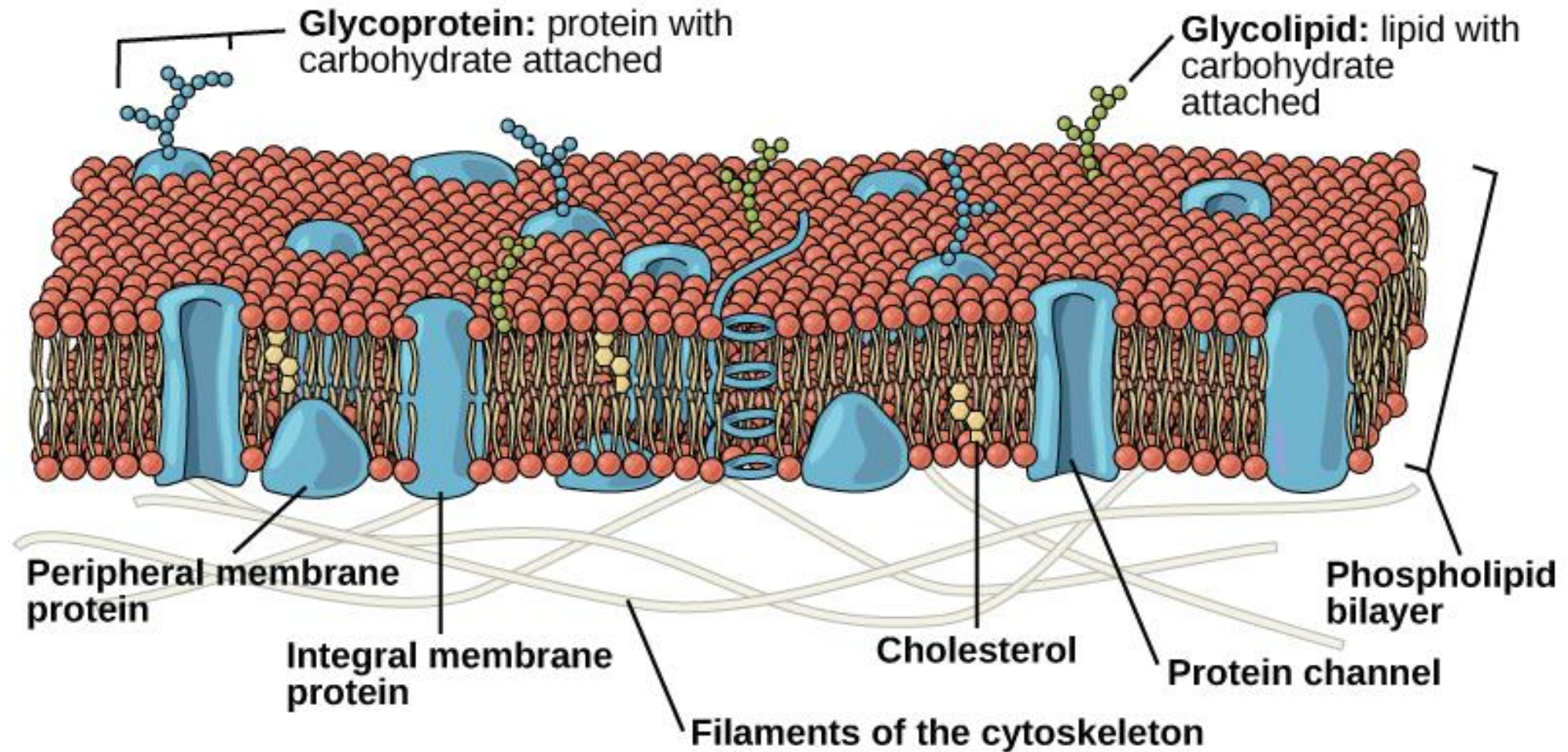
RBC Membrane

- The cell membrane is the biological membrane that separates the interior of all cells from the outside environment, with a basic function to protect the cell from its surroundings.
- Also it is involved in a variety of cellular processes such as cell adhesion, ion conductivity, and cell signaling.
- The RBC membrane consists of approximately 8% carbohydrates, 52% proteins, and 40% lipids (phospholipid and cholesterol).
- This membrane structure provides properties essential for physiological cell function such as **deformability and stability** while traversing the circulatory system and specifically the capillary network.

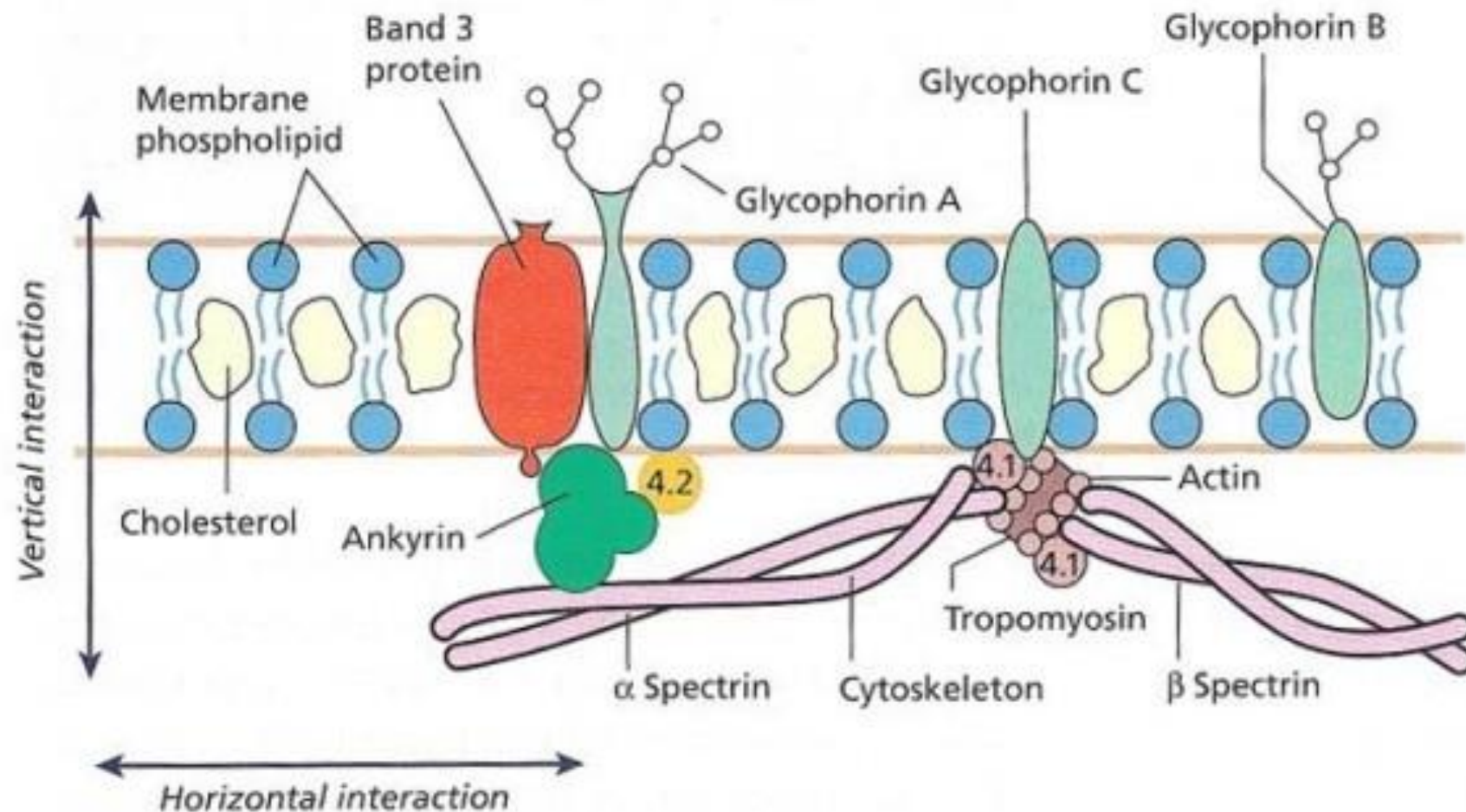
The red blood cell membrane is composed of 3 layers:

- 1) The glycocalyx on the exterior, which is rich in carbohydrates, whose net negative charge prevents microbial attack and protects the RBC from mechanical damage caused by adhesion to neighboring RBCs or the endothelium.
- 2) The lipid bilayer which contains many transmembrane (integral) proteins, besides its lipidic main constituents.
- 3) and the membrane cytoskeleton, a structural network of proteins located on the inner surface (cytoplasmic) of the lipid bilayer. These proteins are called **peripheral proteins**.

RBC membrane



The red cell membrane structure



RBC membrane lipids (40 %):

- Responsible for the bilayer of the plasma membrane.
- Composed of equal parts of phospholipids and cholesterol.
- **Phospholipid:** presents as a bilayer interspersed with molecules of cholesterol, glycolipid, and small quantities of free fatty acids.
- **Cholesterol:** found equally on both layers of the RBC membrane lipid. There is a continual exchange with plasma cholesterol so it is affected by changes in body lipid transport.

➤ **Cholesterol affects membrane fluidity**, if excess then the membrane becomes more viscous and less fluid, which decreases membrane deformability and changes in RBC morphology leading to a decreased survival rate.

➤ **Cholesterol confers tensile strength to the lipid bilayer**

Proteins (52%):

- Although cholesterol and phospholipids constitute the principal RBC membrane structure, transmembrane (integral) and cytoskeletal (skeletal, peripheral) proteins make 52% of the membrane structure by mass
- There are currently more than 50 known membrane proteins, which can exist in a few hundred up to a million copies per erythrocyte.
- Approximately 25 of these membrane proteins carry the various blood group antigens, such as the A, B, and Rh antigens, among many others.
- These membrane proteins can perform a wide diversity of functions, such as transporting ions and molecules across the red cell membrane, adhesion, and interaction with other cells such as endothelial cells, as signaling receptors, as well as other currently unknown functions.

- **Membrane lattice proteins: Spectrin, actin, Band 3, Ankyrin**
- The **proteins** of the membrane skeleton are **responsible for the deformability, flexibility, and durability of the red blood cell**, enabling it to squeeze through capillaries less than half the diameter of the erythrocyte and recover the discoid shape as soon as these cells stop receiving compressive forces, similarly to an object made of rubber.
- Disorder in any of these protein membranes is associated with a pathological condition, such as hereditary spherocytosis (HS) and hereditary elliptocytosis (HE).

A. Band 3:

- The most abundant transmembrane protein.
- An important anion transporter (HCO_3^- and Cl^-).
- An important structural component of the erythrocyte cell membrane.
- Makes up to 25% of the cell membrane surface.
- Each red cell contains approximately one million copies.

B. Glycophorins:

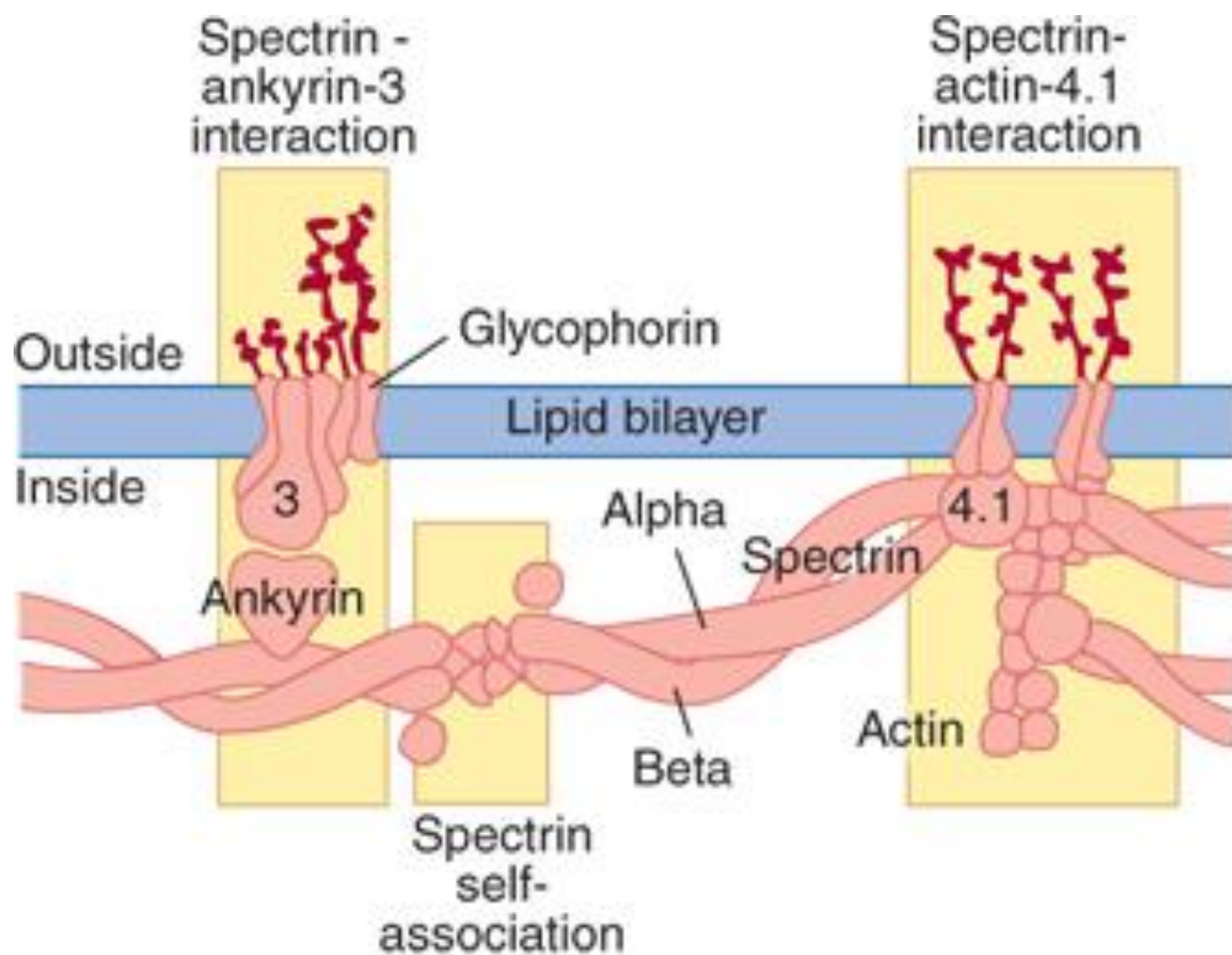
- They are glycoprotein which contains 60% carbohydrate and most of the membrane sialic acid which gives RBCs its negative charge.
- Account for 10% of the total membrane protein.
- **Are the principal RBC glycoprotein.**
- **Are the major integral membrane proteins.**

- Their functions are to carry RBCs antigens and as receptors.
- There are three types of glycoporphins: glycoporphin A, glycoporphin B, and glycoporphin C.

C. Spectrin:

- Is **the most abundant** and makes 25% of the total membrane protein and 75% of the peripheral protein.
- Is **the principal peripheral protein**.
- There are two types of spectrin, designated α and β .
- One α spectrin spontaneously pairs with one β spectrin to form a heterodimer pair; two α / β heterodimers then join together to form a tetramer.

- The β spectrin anchors to the phospholipid bilayer at the band 3 molecule through the **ankyrin** protein, at a structure called the **junctional complex** (vertical connection). The spectrin tetramer anchors to glycophorin through actin and protein band 4.1.
- This connection with the other types of peripheral proteins forms a skeleton of microfilaments that will:
 - I. Strengthen the membrane.
 - II. Protect it from being broken.
 - III. Control biconcave shape and deformability of the cell.

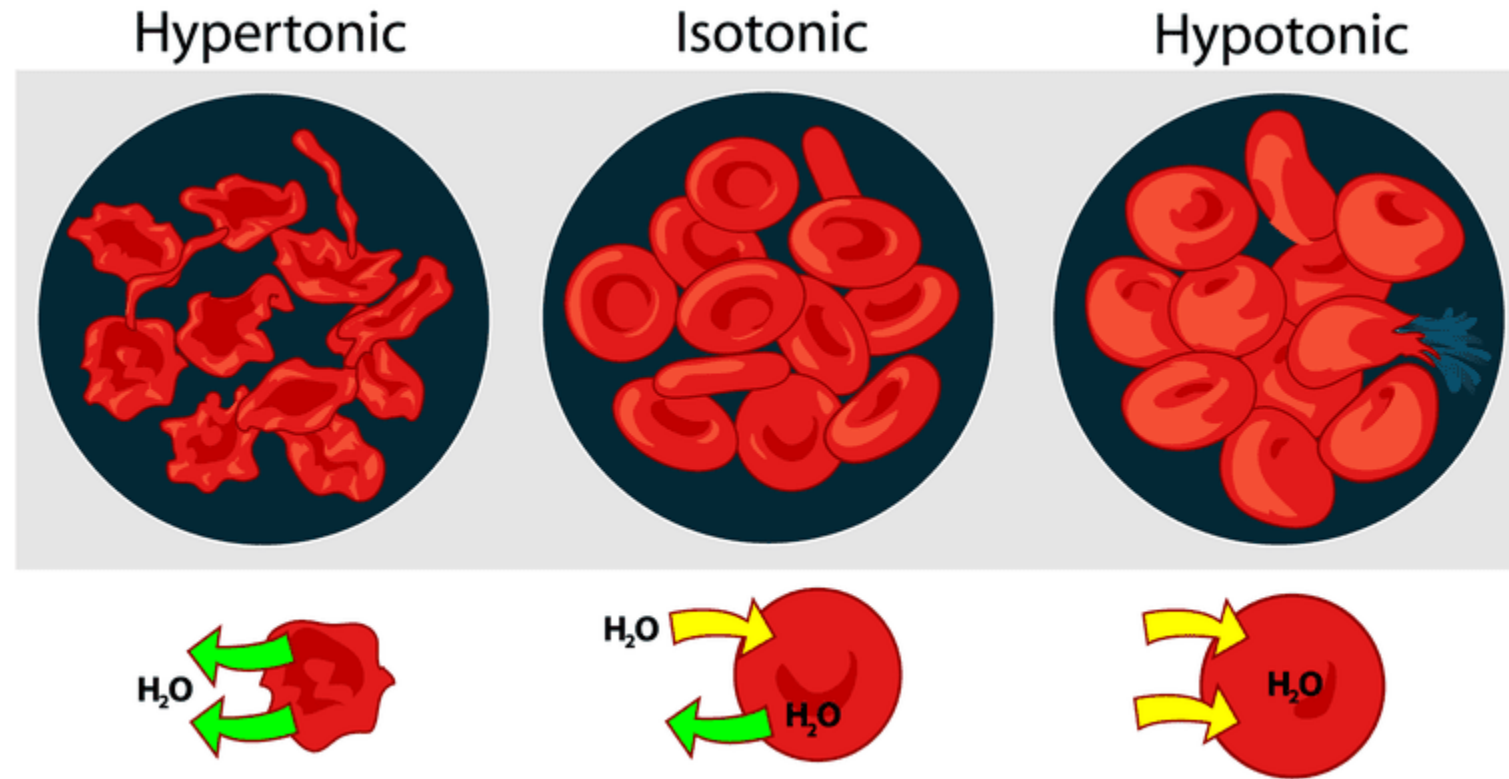


Red blood cell membrane characteristics:

- **Deformability:** loss of ATP → Decrease spectrin phosphorylation → loss of deformability → more rigid.

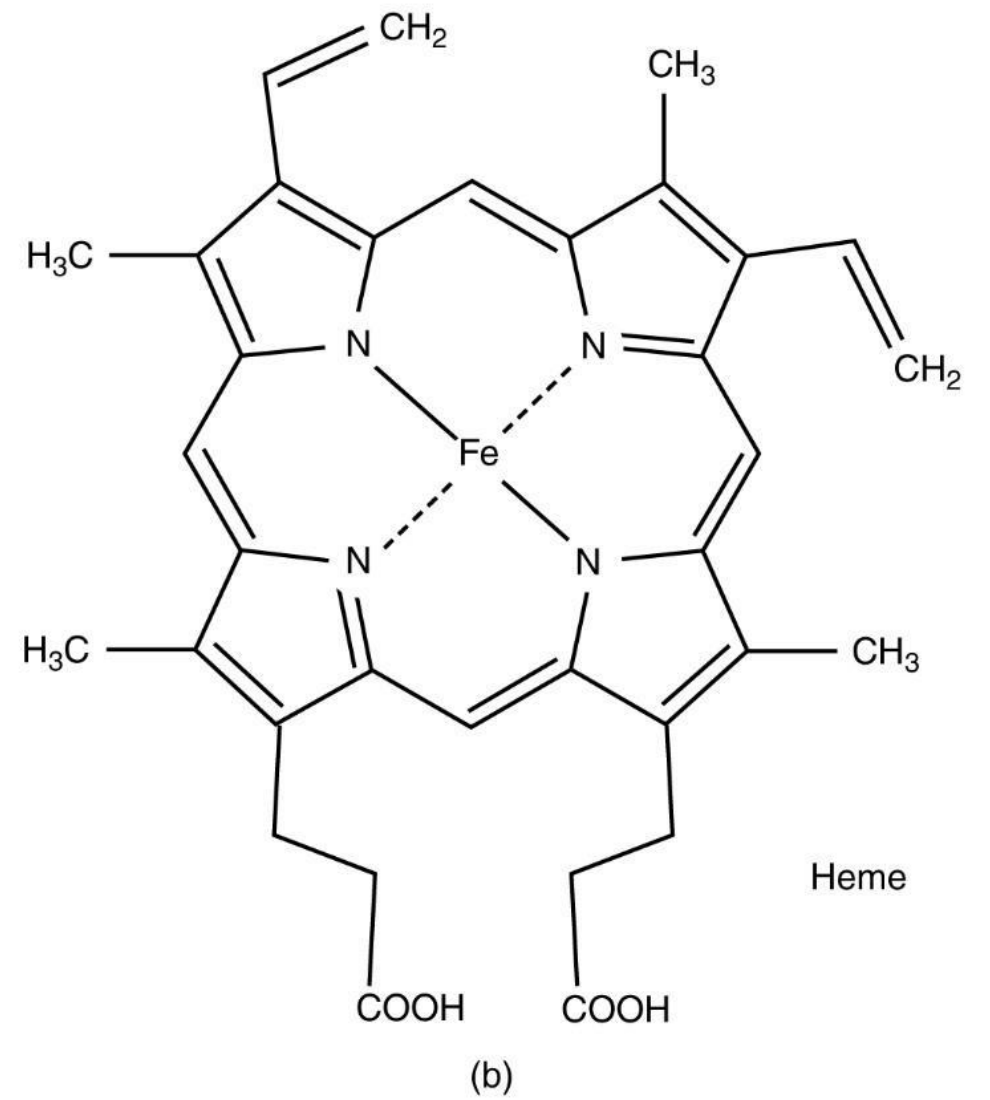
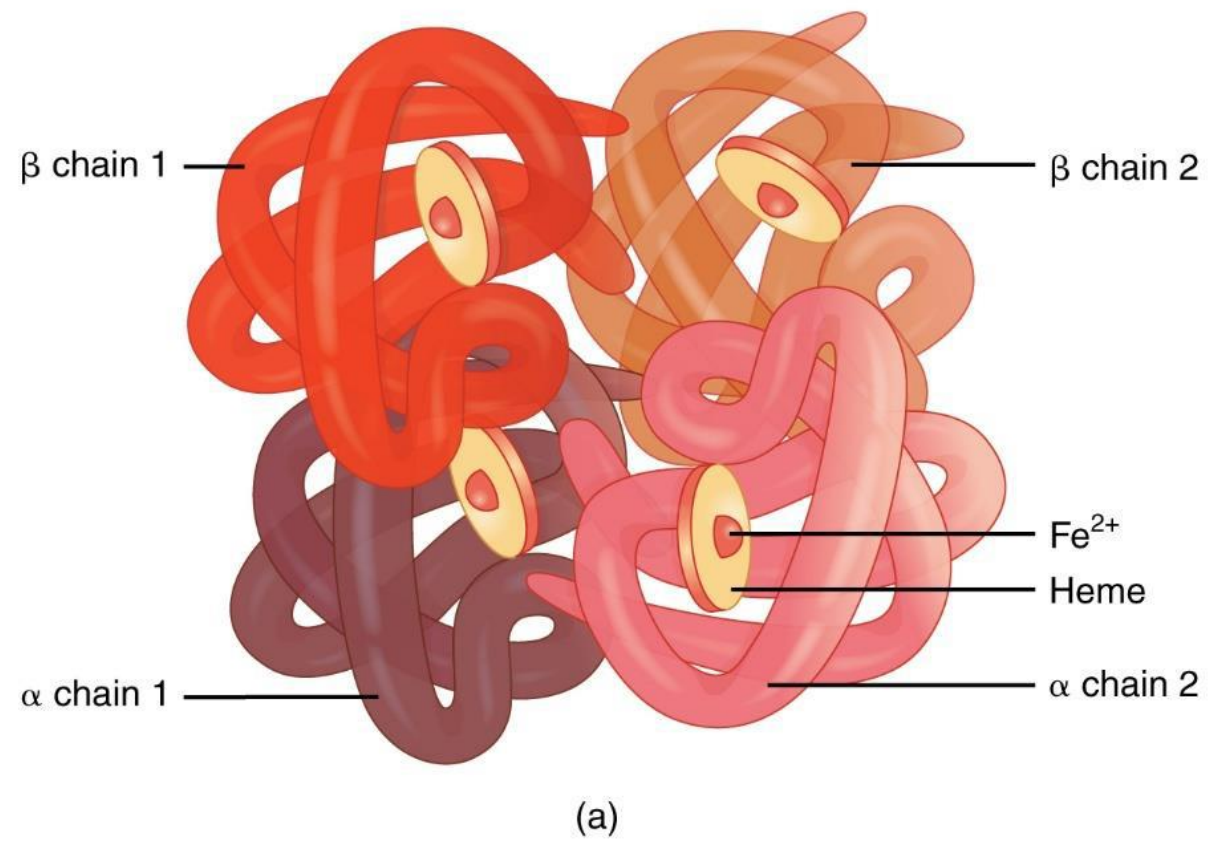
- **Permeability:**
 - 1) It is permeable to water and the anions (HCO_3^- and Cl^-).
 - 2) The RBC membrane is impermeable to cations (Na^+ , K^+ and Ca^{+2}).
 - 3) ATPase-dependent cation pumps (Na, K pump which requires ATP) regulate the concentration of Na^+ and K^+ , maintaining intracellular-to-extracellular ratios of 1:12 for Na^+ and 25:1 for K^+ , which the controlling of their concentrations is important to maintain volume and water homeostasis.

The permeability function of RBC plasma membrane



Hemoglobin

1. It is the iron-containing conjugate metalloprotein, found in the cytoplasm of the RBC and functions in the transport of O_2 from the lungs to tissues and CO_2 from tissues to lungs for exhalation.
2. It is a tetramer protein composed of two parts: heme and globin.
3. It makes up about 95% of the cytoplasmic content of RBCs.
4. The hemoglobin molecule can bind (carry) up to four oxygen molecules.
5. Hemoglobin has an oxygen-binding capacity of 1.34 mL O_2 per gram.
6. The molecule also carries the important regulatory molecule nitric oxide bound to a globin protein thiol group, releasing it at the same time as oxygen.
7. Hemoglobin has a quaternary structure.

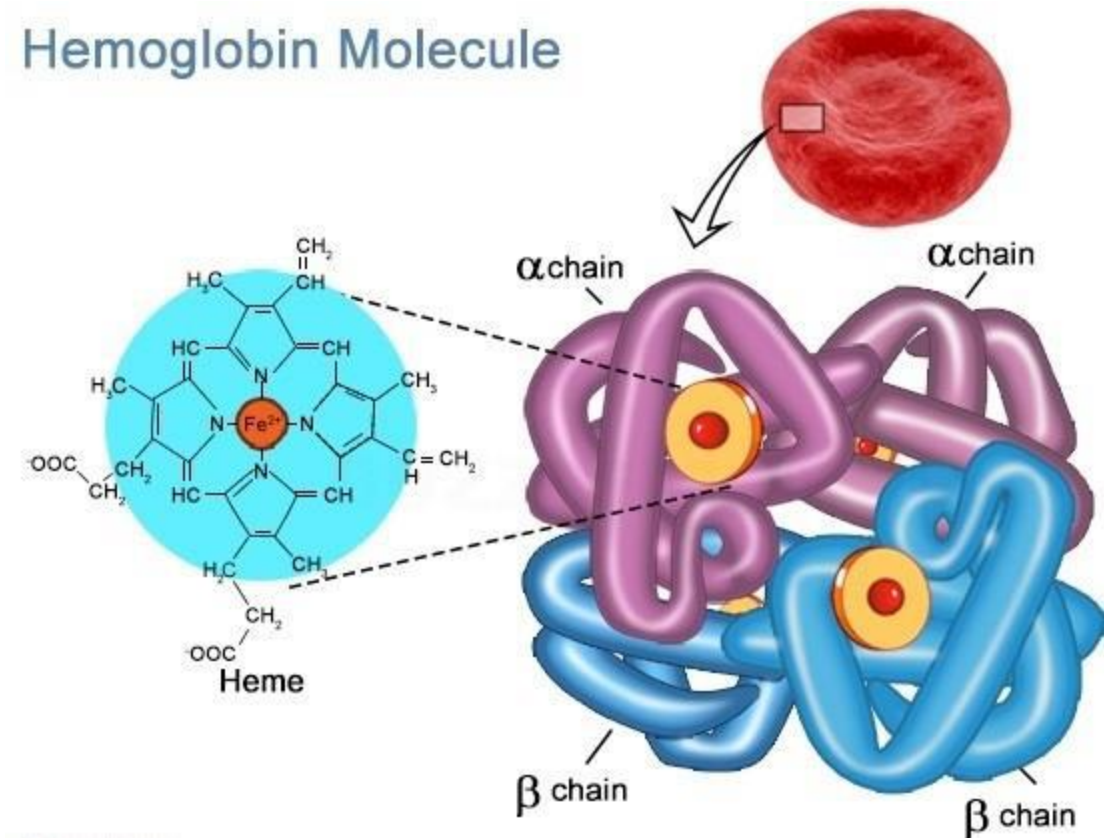


Hemoglobin structure

Hemoglobin (Hb) is a tetrameric protein consists of:

1. Heme group: Four heme groups, each of which contains a protoporphyrin ring and ferrous iron (Fe^{2+}) imbedded in each of the four polypeptide chains.
2. Globin: Tetramer of two different pairs (heterodimer) of polypeptide chains.

Hemoglobin Molecule



➤ Heme: is a chemical group that is composed of two parts:

1. Protoporphyrin IX ring: is a ring composed of four pyrrole rings in which the ferrous iron is located.
2. Ferrous (Fe^{+2}) Iron: is the site of oxygen binding, and coordinates with the four nitrogens in the center of the ring.

➤ It is found in many molecules such as hemoglobin, myoglobin, cytochrome, and catalase.

➤ Its function in the RBC is to carry oxygen from the lungs to the tissues.

➤ It is responsible for the red color of the blood (iron).

➤ The iron should be in the ferrous (Fe^{+2}) state to bind with oxygen, while ferric (Fe^{+3}) iron (metHb) can't bind with oxygen.

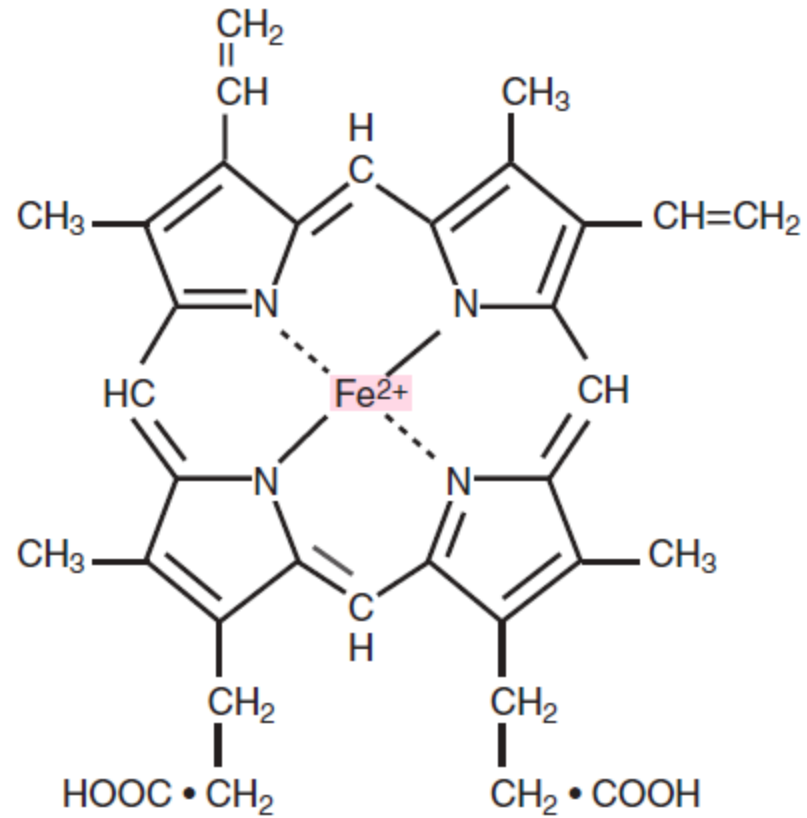
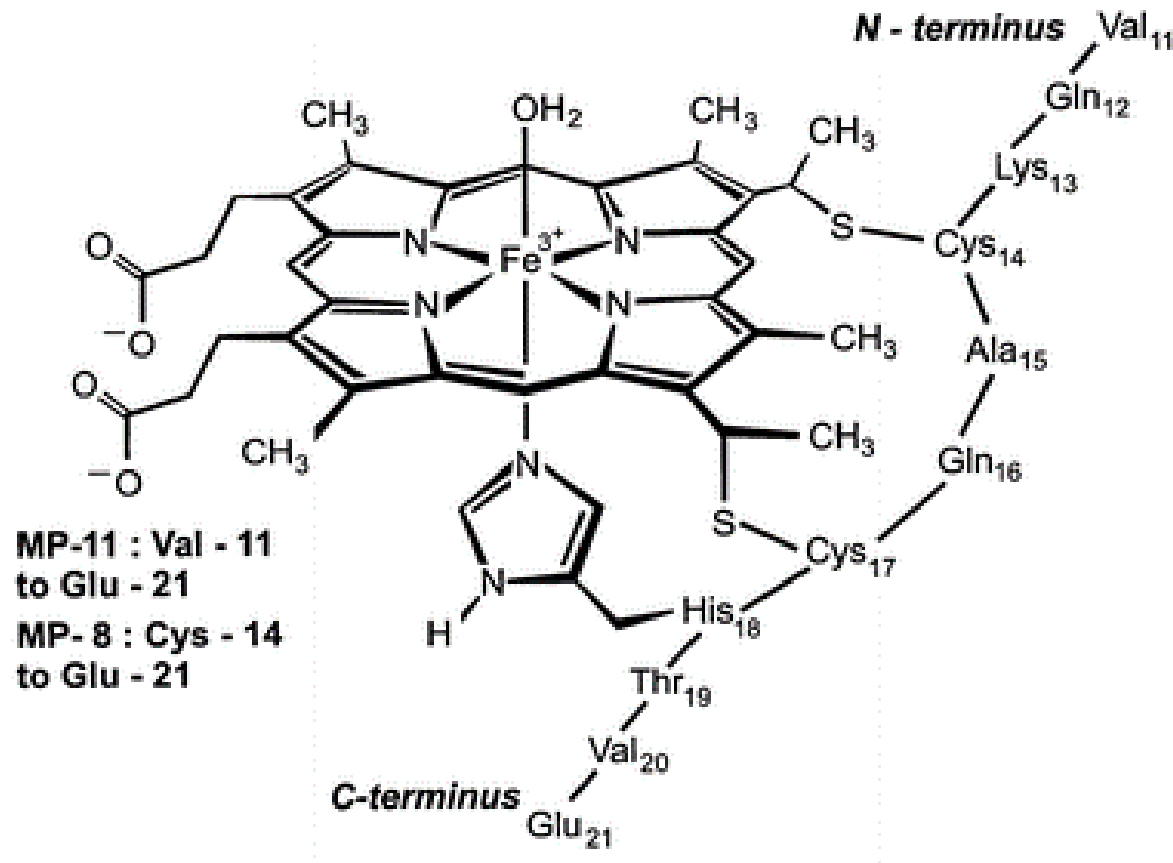


Figure 10-2 Heme is protoporphyrin IX that carries a central ferrous ion (Fe^{2+}).

- Ferrous iron makes 6 bonds, 4 with the N atom of the pyrrole ring, one with the histidine amino acid of the β -globin subunit, and the sixth bond with oxygen.



- Binding of ferrous to the protoporphyrin, globin, and oxygen.

- Globin: is the protein part of the Hb.
- Globin is composed of four polypeptides or subunits (tetramer) that are formed by heterodimers (two different polypeptides).
- The type of polypeptides isn't the same throughout the life.
- The globin polypeptides are composed of 141-146 amino acids.

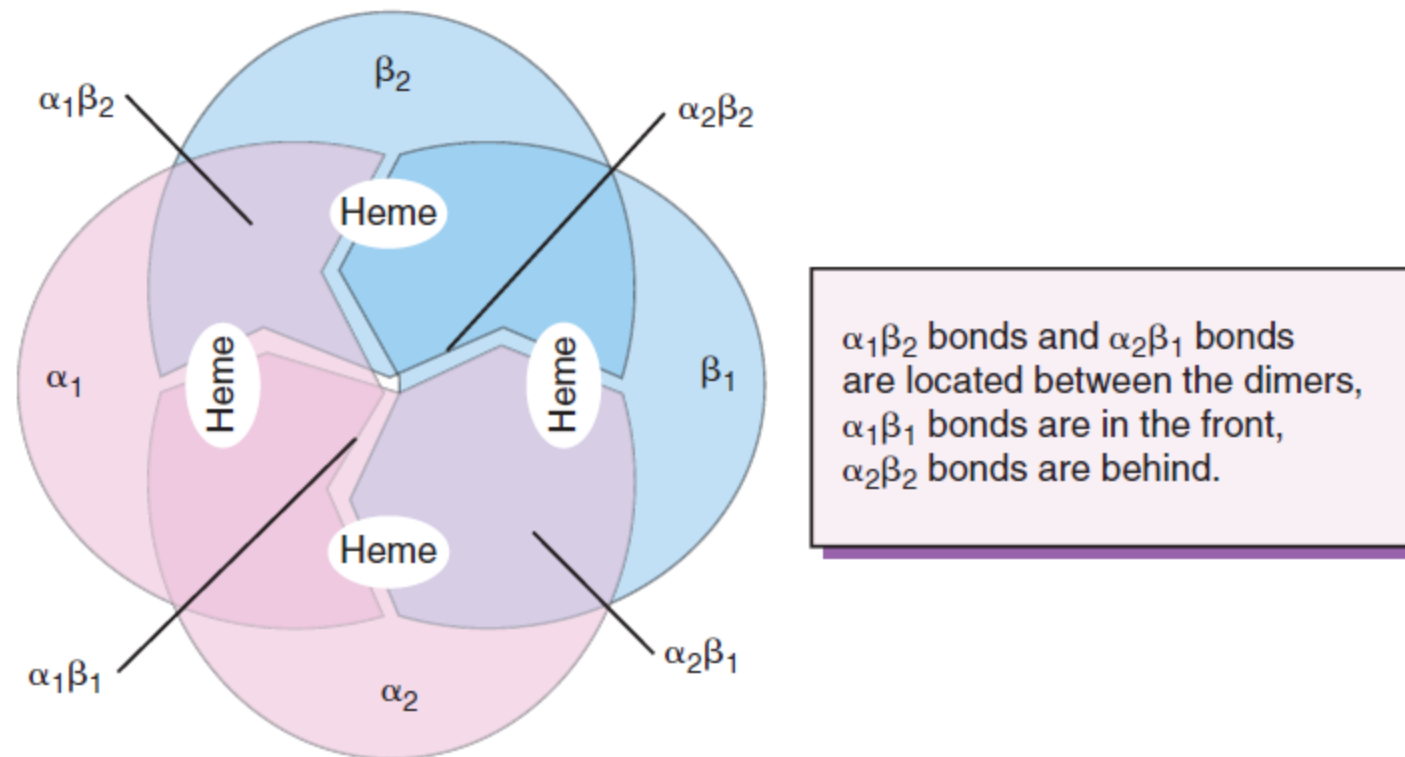


Figure 10-1 Hemoglobin: a tetramer of four globin polypeptide chains, with a heme molecule attached to each chain.

Hemoglobin synthesis

- Normal Hb production depends on three processes:
 1. Adequate iron store and delivery.
 2. Adequate synthesis of protoporphyrin.
 3. Adequate synthesis of globin.

- Hemoglobin (Hb) is synthesized in a complex series of steps.

- The heme part is synthesized in the mitochondria and the cytosol of immature red blood cells, while the globin protein parts are synthesized by ribosomes in the cytosol.

- Heme biosynthesis occurs in the mitochondria and cytoplasm of bone marrow erythrocyte precursors, beginning with the pronormoblast through the reticulocytes.
- As they lose their ribosomes and mitochondria (location of the citric acid cycle), mature erythrocytes can no longer make hemoglobin.
- Heme biosynthesis begins in the mitochondria with the condensation of glycine and succinyl coenzyme A (CoA) catalyzed by aminolevulinate synthase to form aminolevulinic acid (ALA).

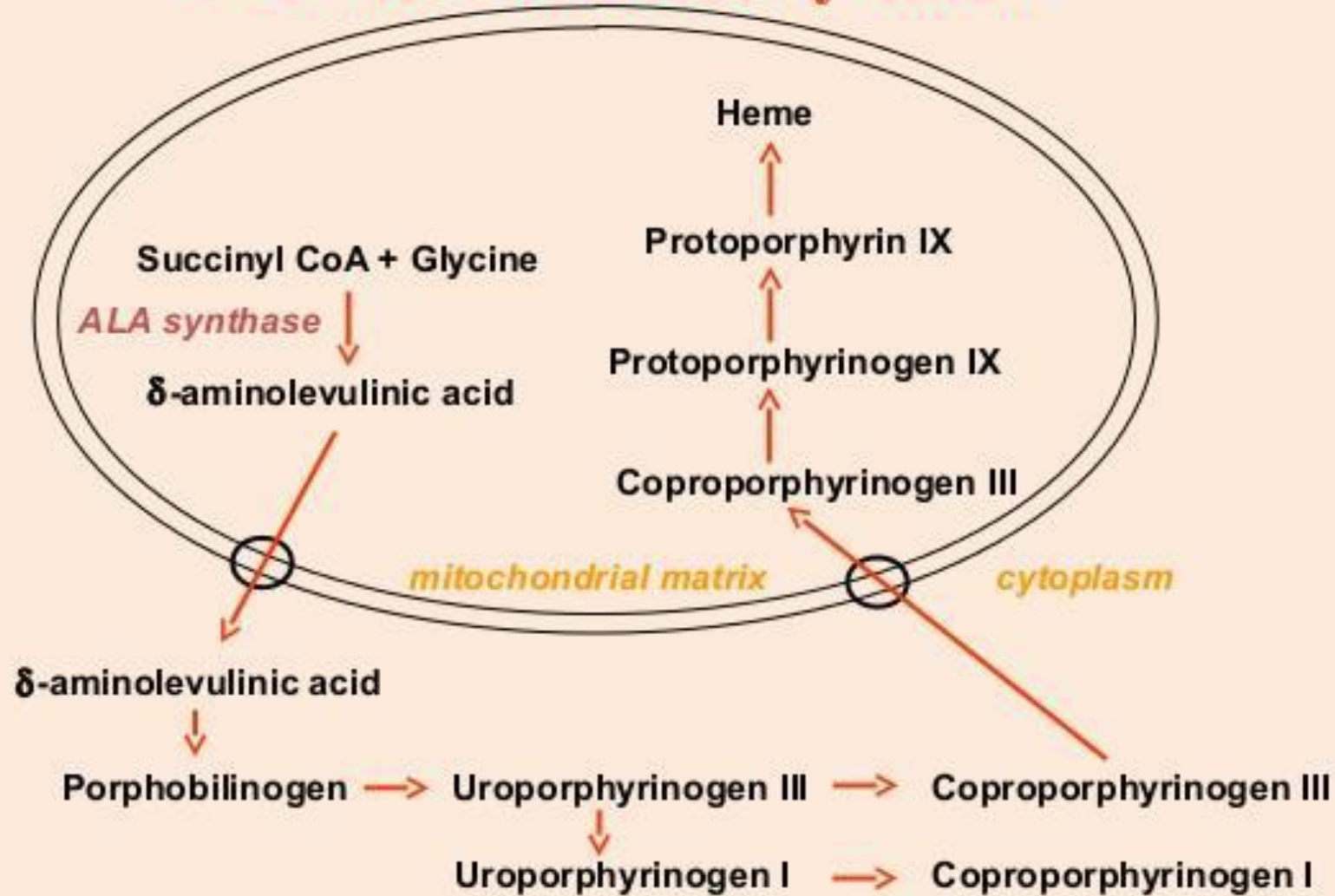
A. Iron delivery and supply:

- Iron in the ferric state (Fe^{+3}) is delivered to the membrane of the RBC precursors by the protein carrier transferrin.
- Iron crosses the membrane to the cytoplasm of the cell and proceeds to the mitochondria, where ferric iron is reduced to ferrous (Fe^{+2}) iron for insertion into the protoporphyrin ring.
- Two-thirds (65%) of the total body iron supply is bound to heme in the Hb molecule.

B. Synthesis of protoporphyrin:

- Begins in the mitochondria by the formation of δ -ALA (aminolevulinic acid) from the amino acid glycine and succinyl coenzyme A (CoA). This reaction is mediated by enzyme, δ -ALA synthetase which is stimulated by EPO and requires the cofactor pyridoxal phosphate (vit B6).
- The total process requires a chain of 8 enzymatic reactions.
- Protoporphyrin IX combines with ferrous iron in the cytoplasm to yield heme.
- Heme leaves the mitochondria and is joined to the globin chains in the cytoplasm.

Overview of Heme Synthesis

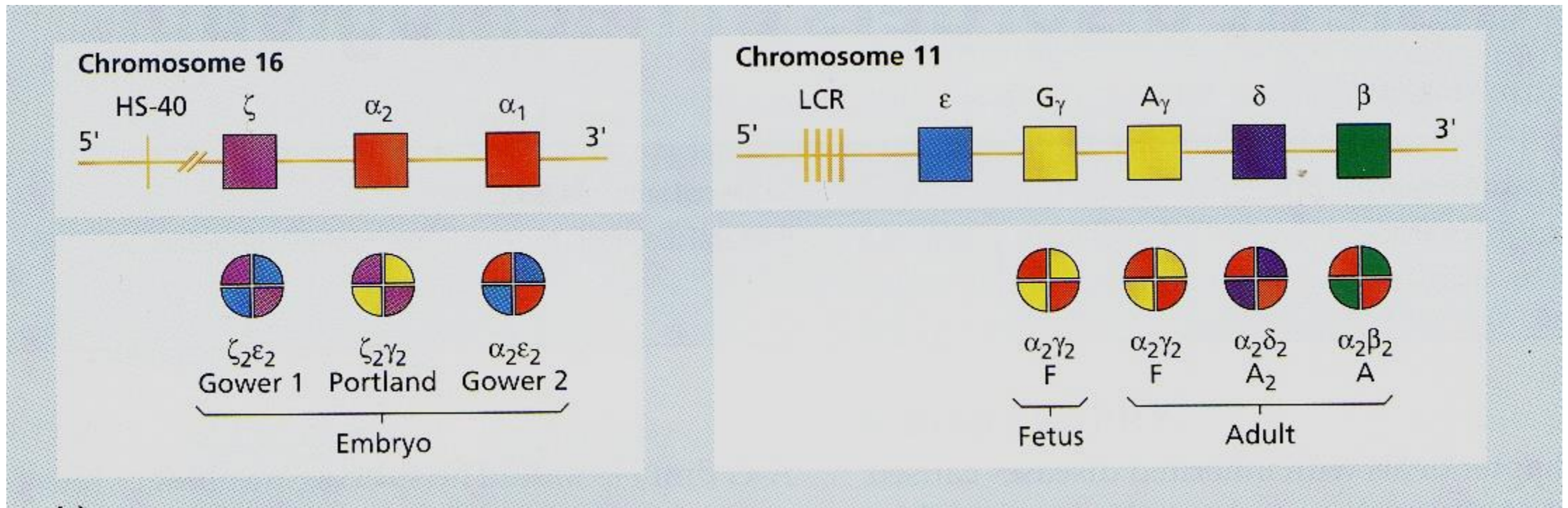


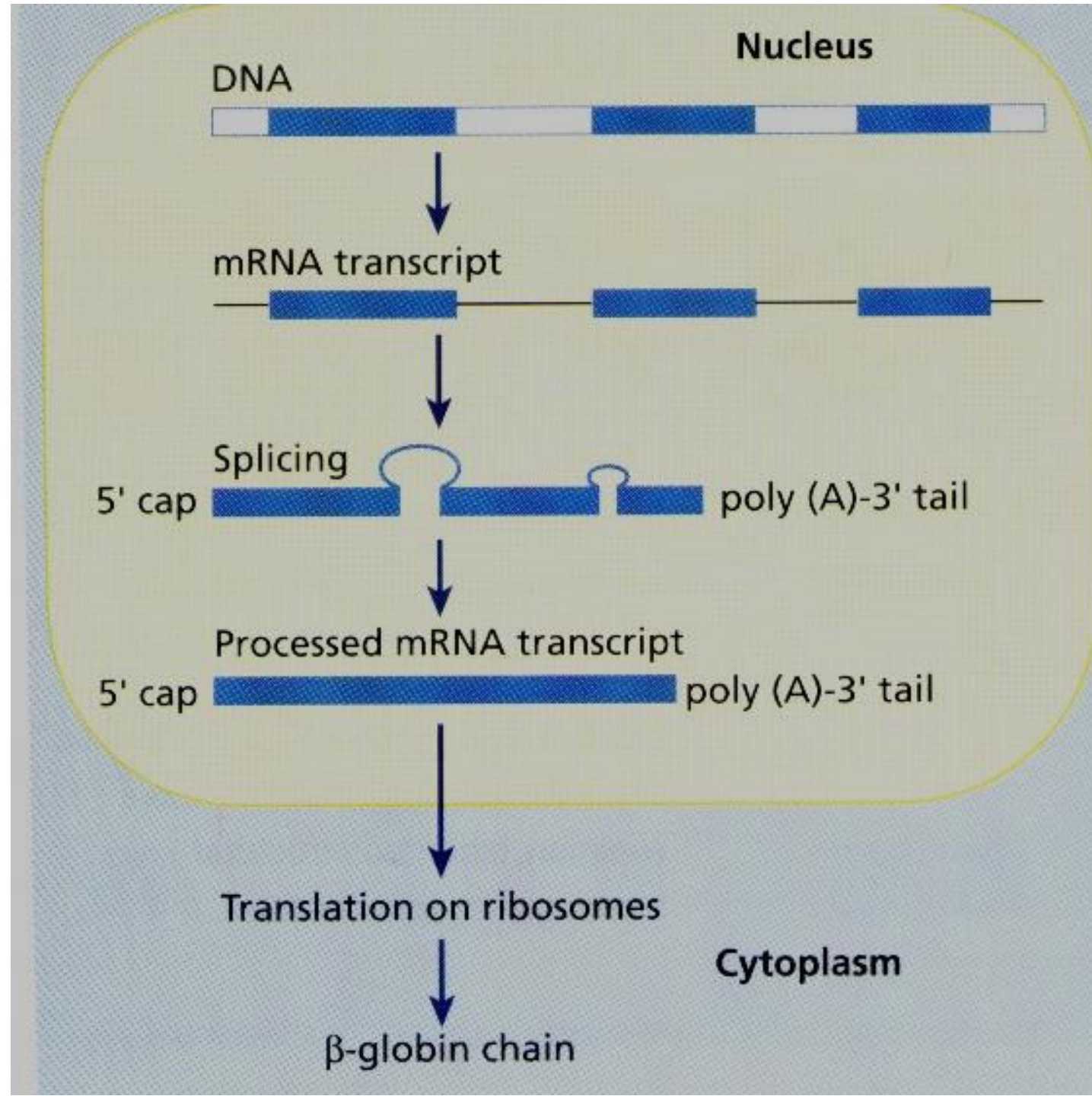
Globin synthesis

- Since globin is a protein, it is encoded by genes located on certain chromosomes within the genetic material and these genes are transcribed and translated to make proteins.
- The translation process, the conversion of mRNA to protein, is achieved by the ribosomes.
- The ribosomes in the cytoplasm of the RBC are responsible for the synthesis of globin proteins.
- The α -like globin polypeptide is composed of 141 amino acids and β -like globin polypeptide is composed of 146 amino acids.

- Production of globin takes place from the proerythroblast (pronormoblast) to the reticulocyte in the bone marrow.
- Even after the loss of the nucleus, residual ribosomal RNA allows further synthesis of globin until the reticulocyte loses its RNA soon after entering the blood circulation.
- After synthesizing at the ribosomes, heme is incorporated into the globin subunit, and then the globin subunits assemble finally resulting in the formation of the Hb molecule (Hb Assembly).

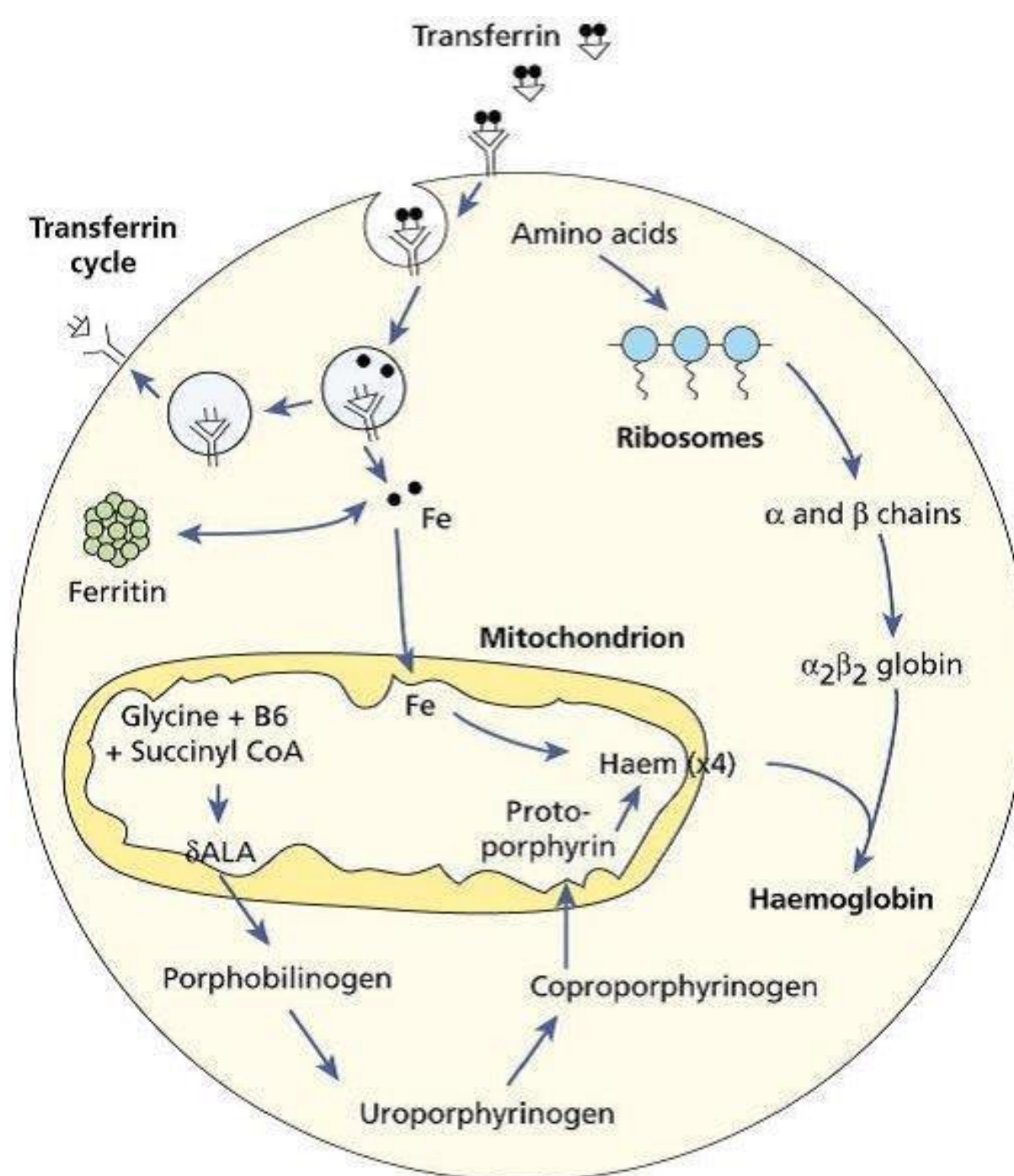
- Eight functional globin chains, arranged in two clusters:
- I. β -cluster (β , γ (2), δ and ϵ globin genes) on chromosome 11.
 - II. α -cluster (α (2) and ζ globin genes) on chromosome 16.





Hemoglobin Assembly

- After their release from ribosomes, each globin chain binds to a heme molecule and then forms a heterodimer.
- Two heterodimers then combine to form a tetramer.
- This completes the hemoglobin molecule.
- Globin synthesis is highly regulated so that there is a balanced production of globin and heme.
- This is critical because an excess of globin chains, protoporphyrin IX, or iron can accumulate and damage the cell, reducing its life span.



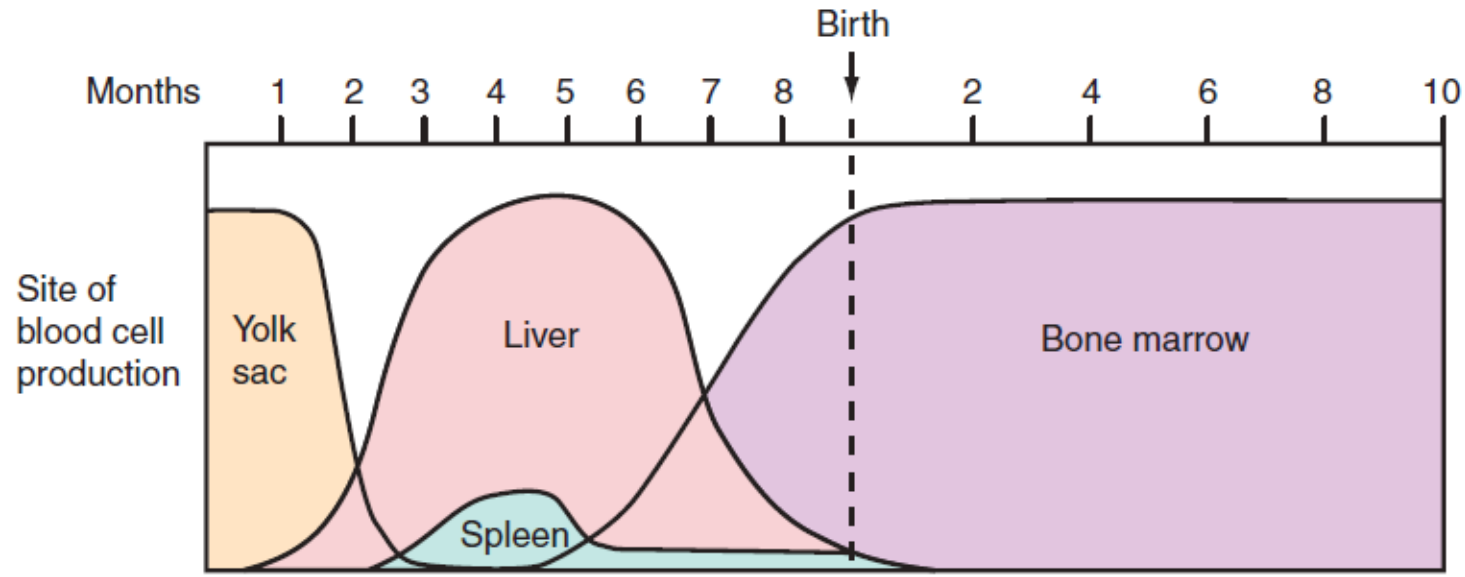
Hemoglobin ontogeny

- Hemoglobin composition differs with prenatal gestation time and postnatal age.
- Hemoglobin changes reflect the sequential activation and inactivation (or switching) of the globin genes.
- **Embryonic Hb:** during the first 3 months of embryonic development.
 - I. Hemoglobin Gower I ($\delta_2\varepsilon_2$)
 - II. Hemoglobin Portland ($\delta_2\gamma_2$)
 - III. Hemoglobin Gower II ($\alpha_2\varepsilon_2$)

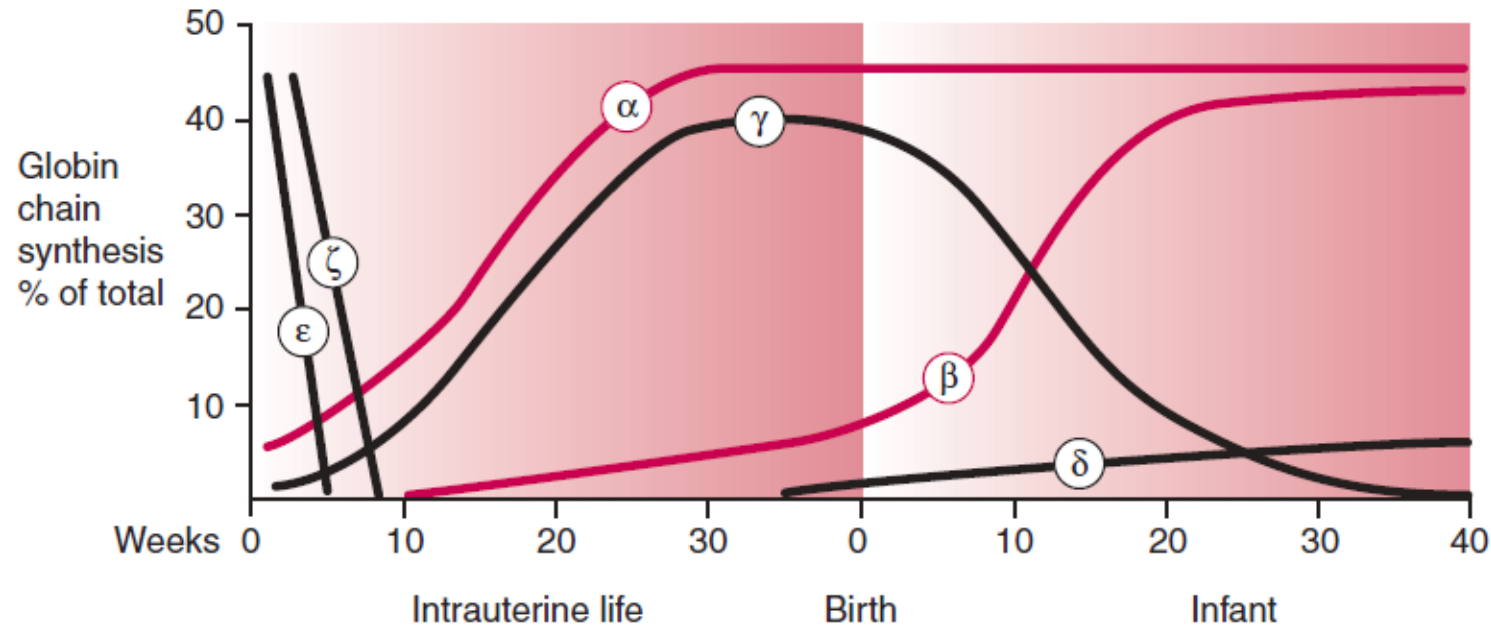
- **Fetal Hb: HbF ($\alpha_2\gamma_2$):** During the second and third trimesters of fetal life and at birth, Hb F ($\alpha_2\gamma_2$) is the predominant hemoglobin.
- By 6 months after birth, a switch from γ to β chain takes place producing HbA $\rightarrow \alpha_2\beta_2$.
- By 6 months of age and through adulthood, Hb A ($\alpha_2\beta_2$) is the predominant hemoglobin, with small amounts of Hb A2 ($\alpha_2\delta_2$) and Hb F.

- **Adult Hb:** HbA ($\alpha_2\beta_2$), HbA2 ($\alpha_2\delta_2$) and HbF ($\alpha_2\gamma_2$).

- Normal adult RBCs contain:
 - I. HbA $\rightarrow$ 96 - 98 %
 - II. HbA2 $\rightarrow$ 1.5 - 3.5 %
 - III. HbF $\rightarrow$ 0 – 2 %



A



B

Figure 10-6 Timeline of globin chain production from intrauterine life to adulthood. See also [Table 10-2](#).

TABLE 10-2 Normal Hemoglobins

Stage	Globin Chain	Hemoglobin
Intrauterine		
Early embryogenesis (product of yolk sac erythroblasts)	$\zeta_2 + \epsilon_2$	Gower-1
	$\alpha_2 + \epsilon_2$	Gower-2
	$\zeta_2 + \gamma_2$	Portland
Begins in early embryogenesis; peaks during third trimester and begins to decline just before birth	$\alpha_2 + \gamma_2$	F
Birth		
	$\alpha_2 + \gamma_2$	F, 60% to 90%
	$\alpha_2 + \beta_2$	A, 10% to 40%
Two Years through Adulthood		
	$\alpha_2 + \gamma_2$	F, 1% to 2%
	$\alpha_2 + \delta_2$	A ₂ , <3.5%
	$\alpha_2 + \beta_2$	A, >95%

- Variations in amino acid sequences give rise to different types of polypeptide chains.
- Each chain is designated by a Greek letter.

TABLE 10-1 Globin Chains

Symbol	Name	Number of Amino Acids
α	Alpha	141
β	Beta	146
γ_A	Gamma A	146 (position 136: alanine)
γ_G	Gamma G	146 (position 136: glycine)
δ	Delta	146
ϵ	Epsilon	146
ζ	Zeta	141

Hemoglobin Function

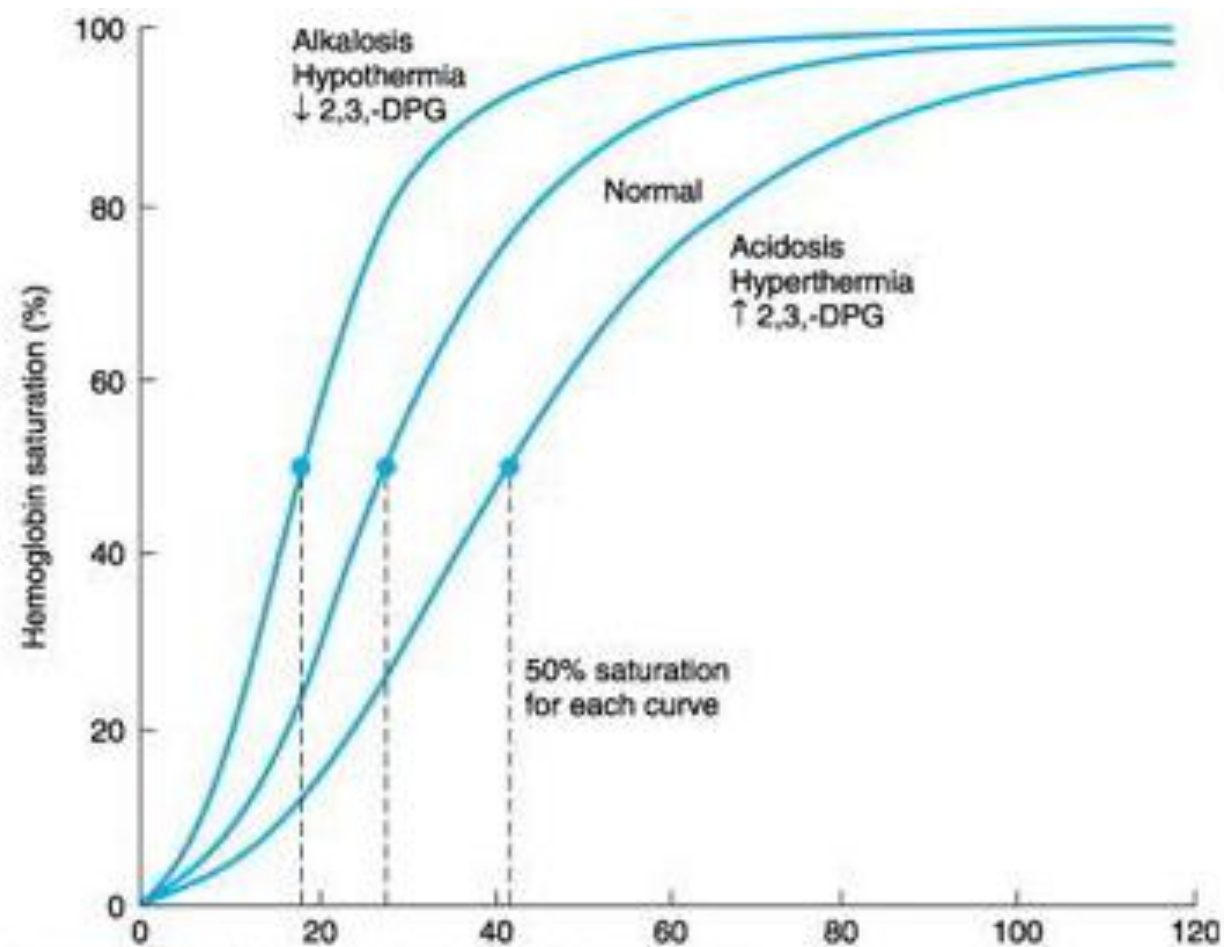
- Each red blood cell (RBC) contains approximately 250 million hemoglobin molecules.
- The main function of hemoglobin is to transport O_2 from the lungs to the tissues and then transport CO_2 back from the tissues to the lungs.
- One hemoglobin molecule can transport up to 4 oxygen molecules.
- Hemoglobin has an oxygen-binding capacity of 1.34 mL O_2 per gram, which increases the total blood oxygen capacity seventy-fold compared to dissolved oxygen in the blood.

- There are two forms of hemoglobin: oxyhemoglobin, which is saturated with oxygen molecules, and deoxyhemoglobin, which is desaturated with oxygen molecules.
- Oxyhemoglobin has a higher affinity for oxygen than deoxyhemoglobin, and deoxyhemoglobin has a higher affinity for CO_2 than oxyhemoglobin.
- Oxygen binds to oxyhemoglobin in the lungs and is then transported through the bloodstream until it reaches the tissues.

- In exchange, the deoxyhemoglobin picks up 2 protons and 2 molecules of CO_2 and returns to the lungs, where the CO_2 is released through exhalation.
- Oxygen binds to hemoglobin with high affinity in an oxygen-rich environment and leaves hemoglobin in an environment where there is not enough oxygen.
- The ability of hemoglobin to take up oxygen molecules in the lungs and then release them in the tissues is regulated by several factors both within the hemoglobin molecule itself and through external chemical factors.

- One of the biggest regulators of the oxygen affinity of the hemoglobin is the presence of oxygen itself. In the lungs where the oxygen levels are high, the hemoglobin has a higher affinity for oxygen.
- The affinity of hemoglobin for oxygen depends on the partial pressure of oxygen (PO_2), which is the amount of oxygen needed to saturate 50% of hemoglobin, called the P50 value.
- The relationship between the affinity of Hb and PO_2 is described by the oxygen dissociation curve of hemoglobin.
- P50 of Hb A, the PO_2 at which 50% of Hb is saturated with oxygen, is 27 mm Hg.
- The reference interval for arterial oxygen saturation is 96% to 100%.

- The Hb dissociation curve may shift to the left ($P_{50} > 27$ mm Hg) or to the right ($P_{50} < 27$ mm Hg).
- Factors that shift the curve to the **left** include: alkalosis (high pH), decreased temperature, decreased PCO_2 , and decreased 2,3 DPG.
- Factors that shift the curve to the **right** include: acidosis (low pH), increased temperature, increased PCO_2 , and increased 2,3 DPG.



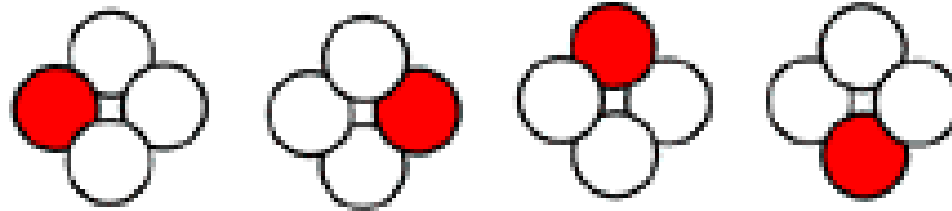
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The effects of changes in acid-base status, body temperature, and 2,3-DPG concentration on the hemoglobin-oxygen **dissociation** curve.

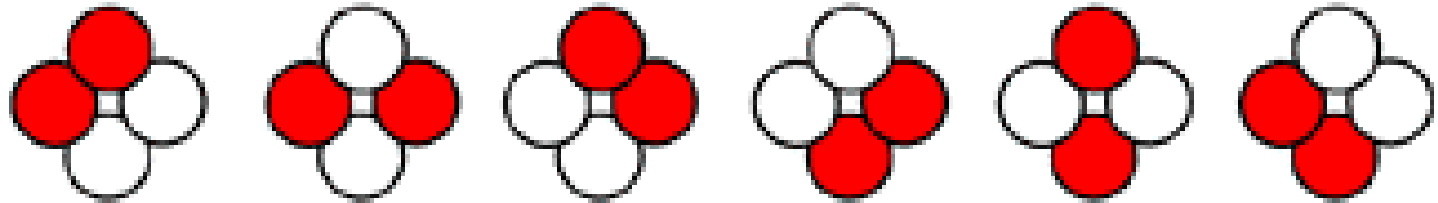
- After the oxyhemoglobin binds one molecule of oxygen, its affinity for oxygen increases until the hemoglobin is fully saturated.
- The deoxyhemoglobin has a lower affinity for oxygen. Thus, the loss of one oxygen molecule from the deoxyhemoglobin lowers the affinity for the remaining oxygen.
- This regulation is known as **cooperativity** (The binding affinity of hemoglobin for oxygen is increased by the oxygen saturation of the molecule, with the first oxygen bound influencing the shape of the binding sites for the next oxygen, in a way favorable for the binding that results in more oxygen binding and makes the binding of successive oxygen molecules more easier) and is essential to the functioning of the hemoglobin because it allows the oxyhemoglobin to carry the maximum amount of oxygen to the tissues and then allows the deoxyhemoglobin to release the maximum amount of oxygen into the tissues.

Cooperativity phenomenon

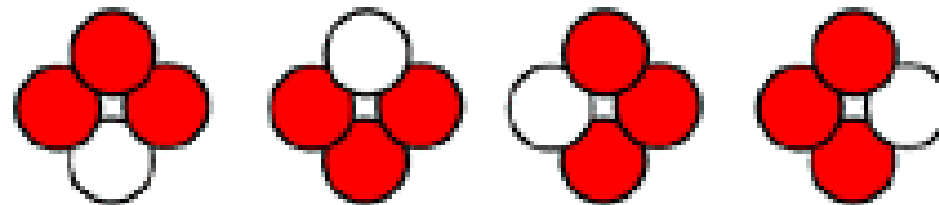
1 O₂ Bound



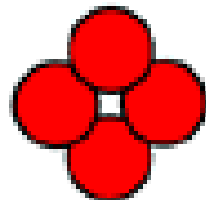
2 O₂ Bound



3 O₂ Bound

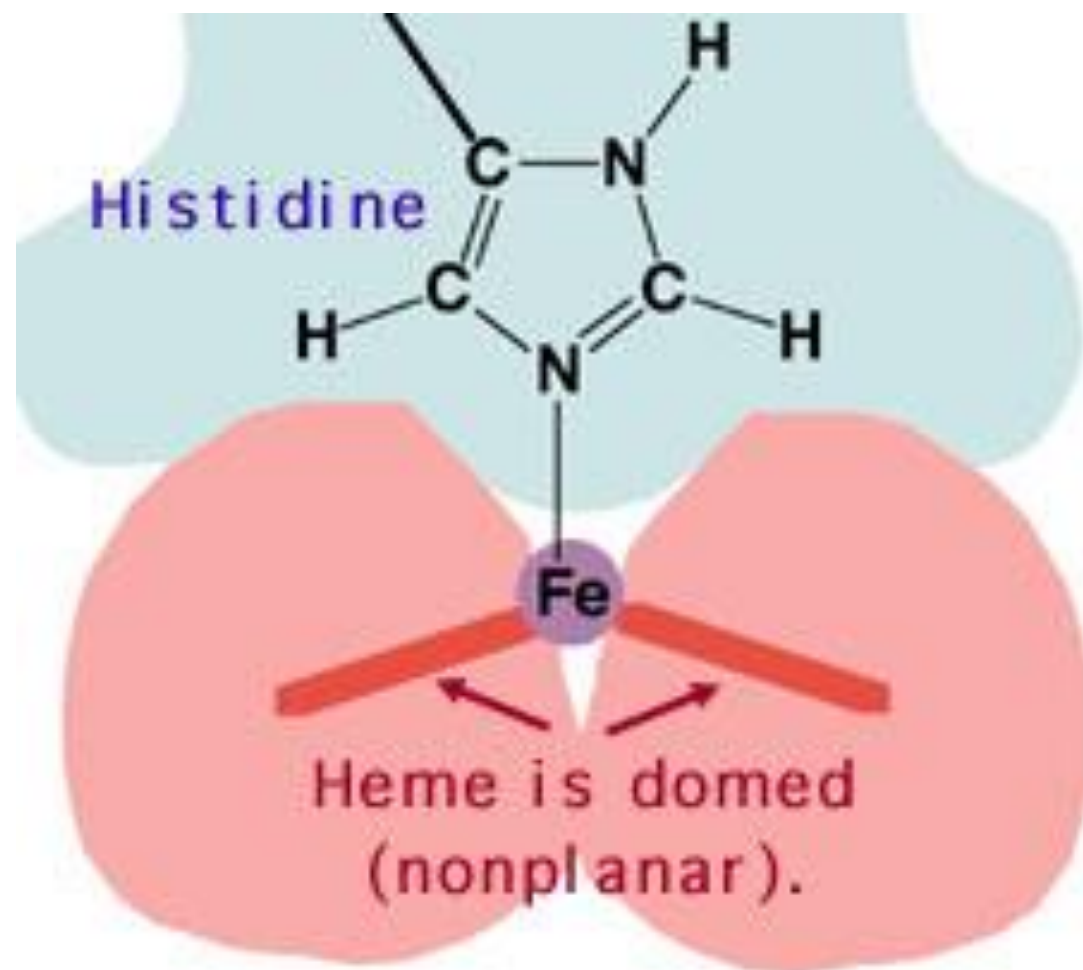


4 O₂ Bound

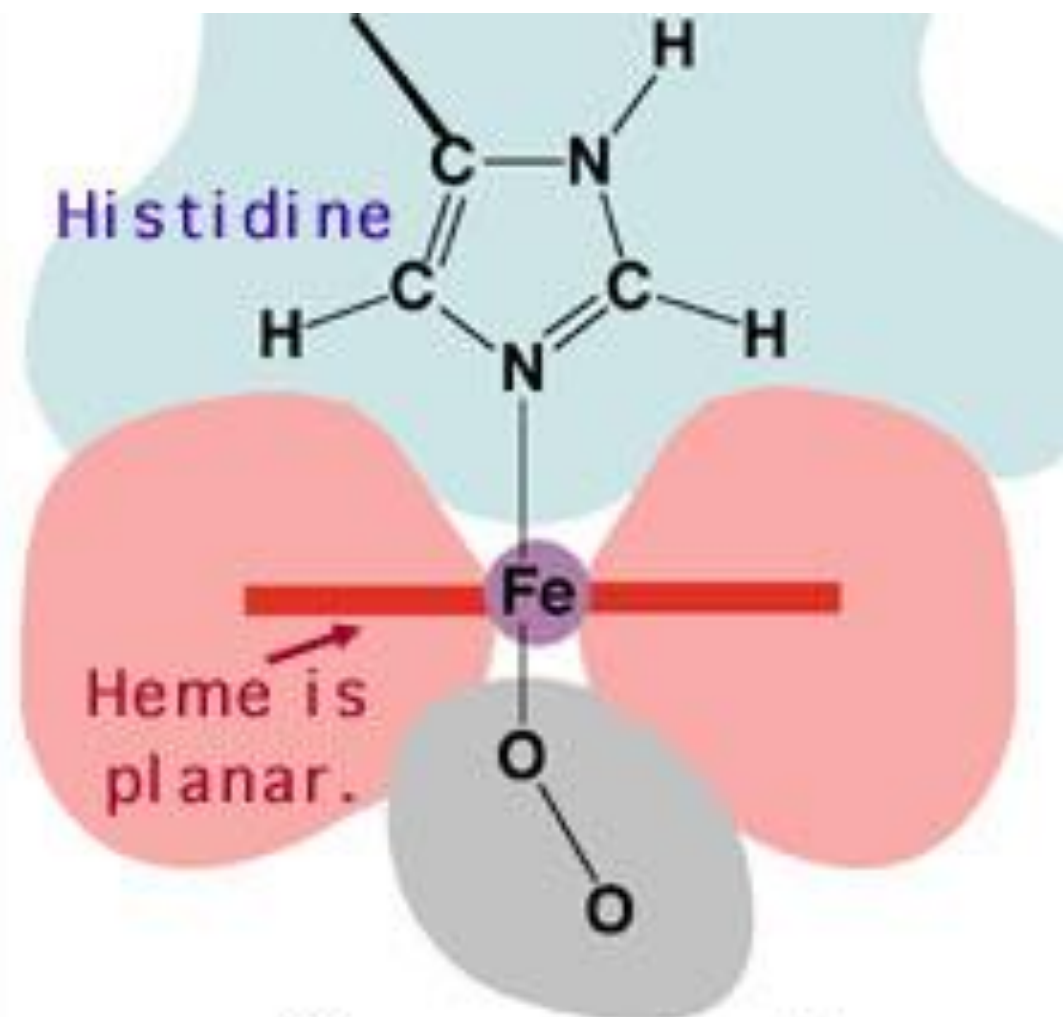


- There are two alternative structures of hemoglobin; the relaxed structure (R) which has a greater oxygen affinity, and the tense structure (T) which has a lower affinity for oxygen.
- In the T structure, the iron ion is pulled out of the plane of the porphyrin ring and becomes less accessible for oxygen to bind to it, thus reducing its affinity to oxygen.
- In the R structure the iron atom is in the plane of the porphyrin ring and is accessible to bind oxygen, thus increasing its oxygen affinity.

- The oxygen affinity of hemoglobin can also be regulated by external chemical factors including pH, carbon dioxide, and 2,3-diphosphoglycerate (2,3 DPG).
- Low pH, high CO₂, and high 2,3 DPG at the level of the tissues favor the T form, which has low oxygen affinity and releases oxygen in the tissues.
- Conversely, a high pH, low CO₂, or low 2,3 DPG favors the R form, which can better bind oxygen.

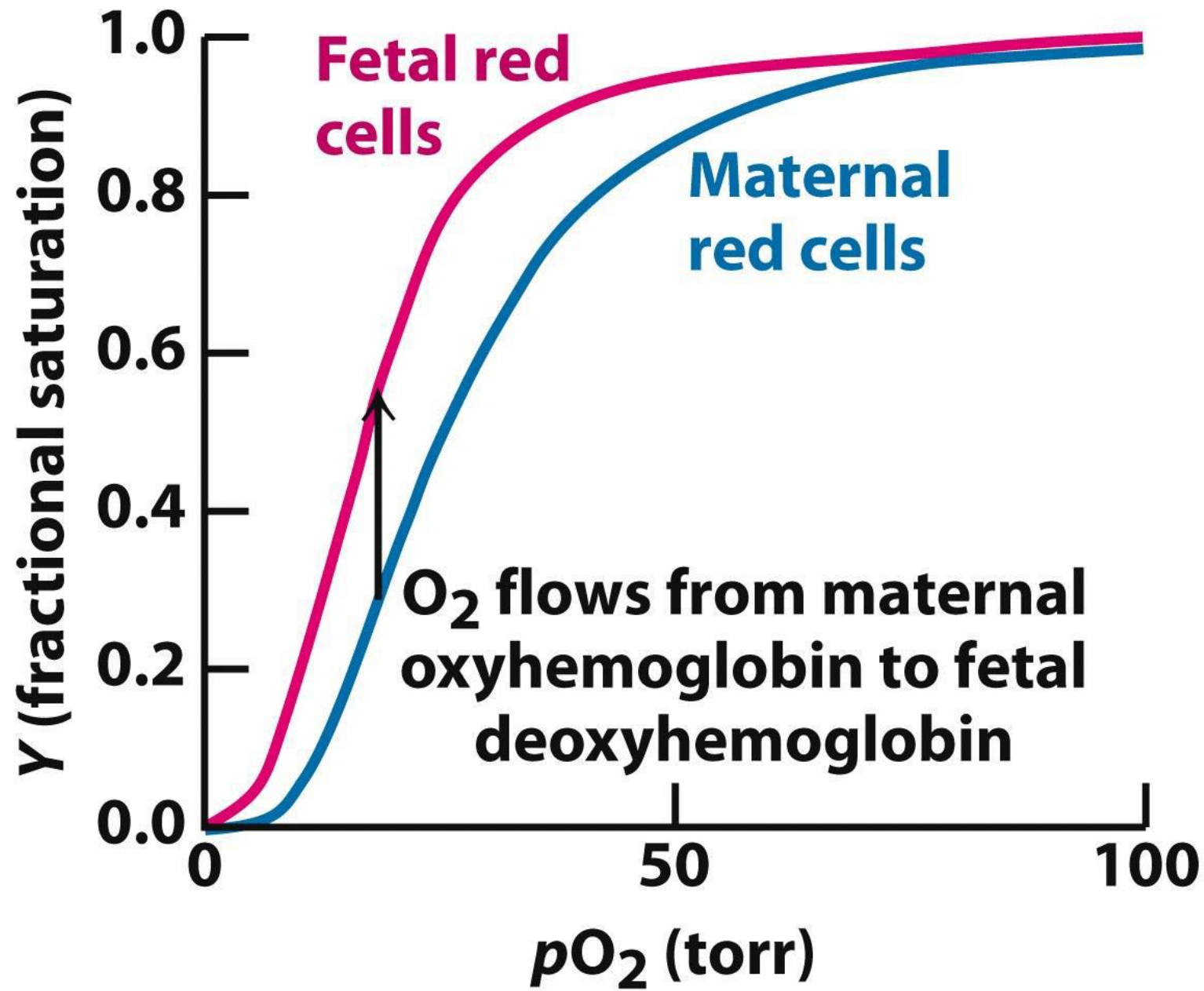


Deoxygenated



Oxygenated

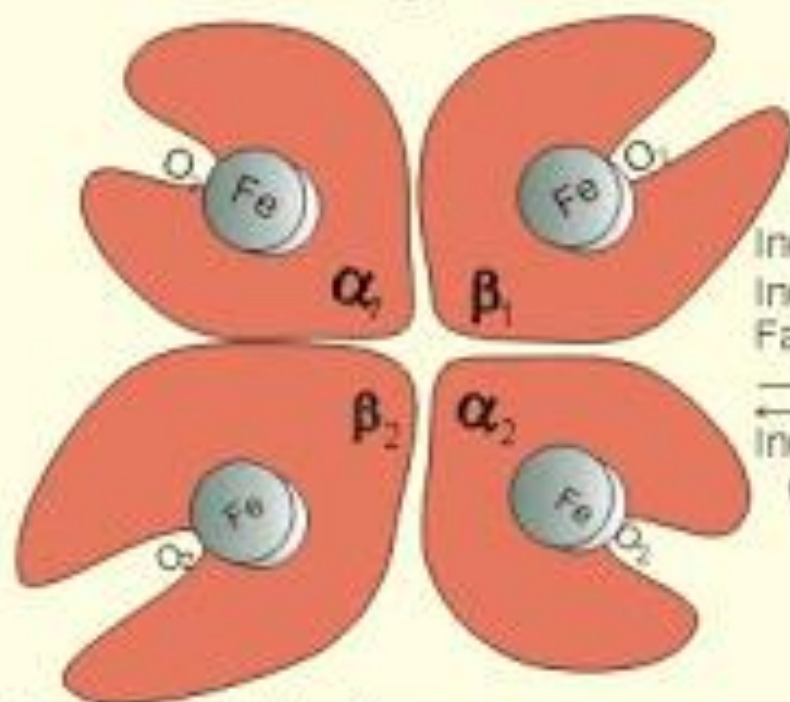
- A variant hemoglobin, called fetal hemoglobin (HbF, $\alpha_2\gamma_2$), is found in the developing fetus and binds oxygen with greater affinity than adult hemoglobin.
- This means that the oxygen binding curve for fetal hemoglobin is left-shifted (i.e., a higher percentage of hemoglobin has oxygen bound to it at lower oxygen tension), in comparison to that of adult hemoglobin.
- As a result, fetal blood in the placenta can take oxygen from maternal blood.
- The P_{50} for Hb F is 19-21 mm Hg.



- The effect of pH on the oxygen dissociation curve is termed the **Bohr effect**.
- If the pH is increased (alkalosis), the curve will shift to the left.
- If the pH is decreased (acidosis), the curve will shift to the right.
- 2,3 DPG will facilitate the release of oxygen from the Hb, thus changing the Hb from the R (oxygenated) to the T (deoxygenated) state.

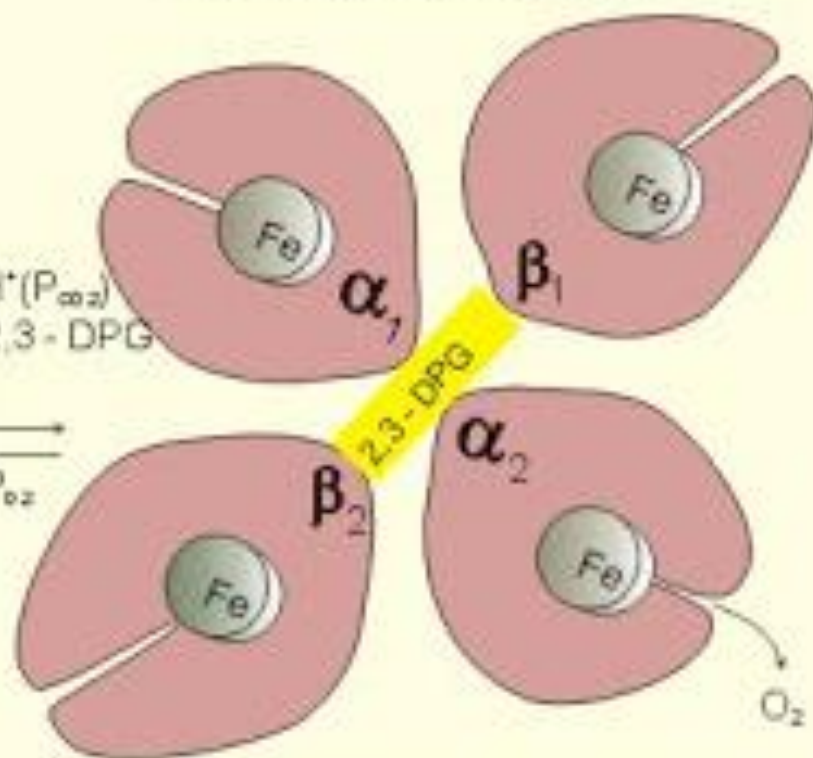
Oxygen Binding and Unloading

Oxyhaemoglobin
Mol weight: 64 460



Relaxed binding structure

Deoxyhaemoglobin



Tight binding structure

Increasing H^+ (P_{CO_2})
Increasing 2,3 - DPG
Falling P_{O_2}

Increasing P_{O_2}
CO

General aspect of anemia:

Anemia: reduction in the haemoglobin concentrations below normal.

Table 2.3 Normal adult red cell values.

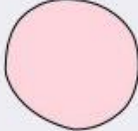
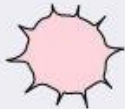
	Male	Female
Haemoglobin (g/dL)	13.5–17.5	11.5–15.5
Haematocrit (PCV) (%)	40–52	36–48
Red cell count ($\times 10^{12}/L$)	4.5–6.5	3.9–5.6
Mean cell haemoglobin (MCH) (pg)	27–34	
Mean cell volume (MCV) (fL)	80–95	
Mean cell haemoglobin concentration (g/dL)	30–35	
Reticulocyte count ($\times 10^9/L$)	50–150	
PCV, packed cell volume.		

Considered anemia if:

- Adult male below 13.5 g/dl
- Adult female below 11.5 g/dl
- Newborn infant below 14 g/dl

Anemia usually but not at all times associated with a reduction in RBC count and PCV level.

Blood film

Red cell abnormality	Causes	Red cell abnormality	Causes
 Normal		 Microspherocyte	Hereditary spherocytosis, autoimmune haemolytic anaemia, septicaemia
 Macrocyte	Liver disease, alcoholism. Oval in megaloblastic anaemia	 Fragments	DIC, microangiopathy, HUS, TTP, burns, cardiac valves
 Target cell	Iron deficiency, liver disease, haemoglobinopathies, post-splenectomy	 Elliptocyte	Hereditary elliptocytosis
 Stomatocyte	Liver disease, alcoholism	 Tear drop poikilocyte	Myelofibrosis, extramedullary haemopoiesis
 Pencil cell	Iron deficiency	 Basket cell	Oxidant damage—e.g. G6PD deficiency, unstable haemoglobin
 Echinocyte	Liver disease, post-splenectomy. storage artefact	 Sickle cell	Sickle cell anaemia
 Acanthocyte	Liver disease, abetalipoproteinaemia, renal failure	 Microcyte	Iron deficiency, haemoglobinopathy

From: *Essential Haematology*, 6th Edn. © A. V. Hoffbrand & P. A. H. Moss.
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